Review Article

Cross-view identity in image-based fruit counting: A systematic review and design space for class-wise inventories

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ABSTRACT

Image-based fruit inventories count fruit from photographs, but a camera moving along a row or around a tree sees most fruit several times, so a sum of detections overstates the true number. Earlier reviews organize this literature by sensor, detector family, or yield-estimation route, and none takes the identity of a fruit across observations as its organizing variable. This systematic mapping review examines how the literature decides which observations belong to the same fruit and how class labels such as maturity are attached to each unique fruit. Reporting follows PRISMA 2020 where its items apply; a Scopus search on 28 September 2026 returned 6,491 records, 5,723 titles were screened, and 1,013 studies published between 2012 and 2026 were included. The 199 studies that combine several observations of the same fruit were coded by association mechanism (no association, statistical correction, appearance matching, temporal tracking, geometric or 3D association, and learned association), acquisition design, and reporting of counts per class, and were summarized with descriptive statistics and a structured narrative synthesis. Temporal tracking dominates recent work, whereas geometric association provides the most direct calibration-free counts when camera pose and depth are reliable. On fruit that look alike, appearance features did not improve tracking over motion alone in the two comparisons made on the same data. Few studies report identity errors or counts per class, and comparisons of mechanisms on the same data are rare. In oil palm, published work concentrates on ripeness classification and bunch detection in single images; cross-view association of bunches on a palm has not been evaluated. The review provides a design space that relates each mechanism to the acquisition conditions it requires and a minimum set of measurements for evaluating duplicate-aware, class-wise inventories.

1. Introduction

A grower who needs the number of ripe fruit on a block wants one record for each physical fruit, with its class. A camera does not supply that record. It supplies images, and in most field protocols the same fruit appears in several of them: in consecutive video frames, in photographs taken from both sides of a row, or in images taken while walking around a single tree. Detection finds fruit in each image. It does not say whether a fruit found in one image is the fruit found in another. The step that makes that decision determines whether a sum of detections is an inventory or an overcount.

The size of the problem is known from studies that measured it directly. In a mango orchard, one image of each side of 16 trees counted 54% of the fruit that were hand-counted, and one image of one side counted 27%; tracking every fruit through a multi-view sequence and triangulating it counted 101.4% (Stein et al., 2016). On 21 mango trees, video tracking recorded 62% of the harvest count and two images per tree recorded 40% (Wang et al., 2019). On a 192-fruit video, the same tracker produced 9.9% double counts and 7.3% missed fruit, which netted to a 2.6% overcount (Wang et al., 2019). The last figure shows why a total count can look correct when the underlying identity decisions are not: a duplicate and a miss cancel.

In oil palm the problem takes a specific form. Plantations estimate output with a bunch census in which staff count and classify bunches on each palm by the number of months until harvest. The SawitMVC dataset photographed 953 palms from four or eight sides and annotated 9,823 unique bunches in four census classes, with links that record which boxes in different images show the same bunch (Indriani et al., 2026). A census built from such photographs needs both the identity decision and a class label for each unique bunch.

From the perspective of counting unique fruit, earlier reviews leave three things open. First, they organize the literature by sensor (Gongal et al., 2015; Fu et al., 2020a), by detector family (Koirala et al., 2019; Huang et al., 2023), by yield-estimation route (Anderson et al., 2021b; He et al., 2022a), or, for oil palm, by ripeness-sensing method (Lai et al., 2023; Goh et al., 2025a); none takes the identity of a fruit across observations as the organizing variable, although several discuss occlusion and extrapolation from partial counts. Second, reviews of tracking, re-identification, and 3D reconstruction outside agriculture provide the mechanisms but not the agricultural evidence. Third, five of the eleven closest reviews do not state how their studies were found, and two more describe themselves as systematic without reporting databases or search strings (Table 6). As a result, a reader cannot tell which association mechanism fits a given acquisition protocol, or what evidence supports it.

To address these issues, this article presents a systematic mapping review of image-based fruit counting published

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between 2012 and 2026. Reporting follows PRISMA 2020 where its items apply to a mapping review. Scopus was the single database, searched on 28 September 2026, and 1,018 studies are included, of which 200 combine several observations of the same fruit.

This review makes three contributions. First, it provides a systematic map of the 1,018 studies, with the search strings reported in full and the screening decisions, the codes, and the log of changes made during verification provided as supplementary material. Second, it defines a design space that separates acquisition, observation, association, attribute assignment, and inventory, and classifies association into six mechanisms (no association, statistical correction, appearance matching, temporal tracking, geometric or 3D association, and learned association), each tied to the conditions it assumes. Third, it synthesizes the 200 multi-observation studies by mechanism, with the comparisons that were made on the same data reported in two tables, followed by the evidence on class attributes, depth, and oil palm.

Accordingly, the review addresses four questions:

(1) Which mechanisms have been used to decide that observations in different images belong to the same fruit, and what acquisition conditions does each assume?

(2) What evidence shows how well each mechanism controls duplicates and misses, and how was that evidence measured?

(3) How are class attributes, mainly maturity, attached to a unique fruit rather than to each detection?

(4) What does the oil-palm imaging literature cover, and what is missing for a bunch-level, class-wise census from images?

The remainder of this article is organized as follows. Section 2 describes the search, screening, and coding. Section 3 sets out the design space and reports how the multi-observation studies are distributed over it. Sections 4 to 6 follow its stages: acquisition (Section 4), association, mechanism by mechanism (Section 5), and class attributes on unique fruit (Section 6). Section 7 examines depth and other sensing modalities, and Section 8 the oil-palm case. Section 9 places the review against earlier reviews, Section 10 draws the gaps and a measurement agenda, Section 11 states the limitations, and Section 12 concludes.

2. Review method

The review is a systematic mapping review with a configurative synthesis (Petersen et al., 2015; Grant and Booth, 2009). It maps a broad body of work and builds a classification from it; it does not pool effect sizes, because the included studies use different crops, references, and metrics. Reporting follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) where the items apply to a mapping review (Page et al., 2021a,b), together with general guidance on reporting literature reviews written for management and transport research (Kraus et al., 2022; Wee and Banister, 2016). The workflow comprised (i) defining the four questions and the scope, (ii) specifying inclusion and exclusion criteria, (iii) searching Scopus, (iv) screening records in two stages, (v) coding the included studies, and (vi) appraising and synthesizing the evidence.

2.1. Data source

Scopus was used as the single bibliographic database because it indexes the journals and proceedings in which agricultural engineering and computer vision work is published, including those of Elsevier, IEEE, Springer, MDPI, and Frontiers. The search was run on 28 September 2026 (UTC).

2.2. Search strategy

The search comprised 13 search strings in nine query sets (Table 1). Five sets target agricultural fruit perception: multi-observation counting, counting in titles, oil-palm imaging, class attributes with counting, and depth or 3D sensing. Two sets retrieve reviews: methods reviews on multi-object tracking (MOT), re-identification (re-ID), multi-view detection, and reconstruction, and agricultural reviews on fruit detection and yield. Two sets retrieve methods papers from outside agriculture; their results were sorted by citation count and only the 25 most cited records of each search were retained, because their role is to identify the transferable mechanisms, not to map those fields. All searches were limited to 2012–2026, and all except the named-methods set (Q9) were also limited to subject areas that publish engineering or agricultural work. The complete search strings are given in Appendix B under the keys shown in Table 1.

2.3. Inclusion and exclusion criteria

Records published from 2012 to 2026 were included if they met at least one of the following: (i) an image-based method for fruit on the plant, or on the ground under the plant for oil-palm loose fruit, that evaluated counting, yield estimation, detection with a class attribute, localization, or sizing; (ii) a review of these topics; or (iii) a methods paper on tracking, re-identification, multi-view association, 3D reconstruction, or object counting outside agriculture. Records were excluded if: (i) the object was not fruit on the plant (for example trees, leaves, or flowers only); (ii) imaging was postharvest or laboratory-only, except oil-palm mill grading, which is part of the case; (iii) yield was modelled without fruit-level detection; (iv) the work addressed only picking or manipulation; (v) fruit were detected without counting or any evaluation beyond detection; (vi) the sensor was not an imaging sensor; or (vii) the record had no English title and abstract.

2.4. Study selection and screening

Records passed through two stages: title screening and eligibility assessment on the title and abstract. The criteria in Section 2.3 were applied at both stages, and the reason for every exclusion at the eligibility stage was recorded (Fig. 1).

Table 1
Scopus query sets and number of records returned and retained.

Query set (key)	Search terms	Returned	Retained
Multi-observation counting (Q1)	Fruit or crop terms AND counting or yield AND multi-view, video, tracking, SfM, SLAM, duplicates	871	871
Counting in titles (Q2)	Fruit AND counting or load in the title	757	757
Oil-palm imaging (Q3)	Oil palm or FFB AND imaging, learning, ripeness, or counting	1,079	1,079
Class attributes with counting (Q4)	Fruit AND ripeness, grade, or class AND counting AND detection	361	361
Depth and 3D sensing (Q5)	Fruit AND RGB-D, stereo, LiDAR, or point cloud AND detection or counting	1,988	1,988
Reviews of association methods (Q6)	Reviews of MOT, re-ID, multi-view detection, SfM, SLAM	236	236
Reviews of fruit detection and yield (Q7)	Reviews of fruit detection, counting, ripeness, yield	1,049	1,049
Methods outside agriculture, five title searches (Q8a–Q8e)	MOT, multi-view, SfM/SLAM, object counting, re-ID in the title	23,532	125
Named foundational methods (Q9)	Titles of named methods (SORT, NeRF, SAM, …)	1,082	25
Total			6,491

Note: Full strings are given in Appendix B under the same keys. For the last two sets only the 25 most cited records of each search were retained. FFB, fresh fruit bunch; SfM, structure from motion; SLAM, simultaneous localization and mapping; MOT, multi-object tracking; re-ID, re-identification; RGB-D, red-green-blue-depth; LiDAR, light detection and ranging.

Duplicates were removed by Scopus record identifier. Proceedings front matter, errata, notes, editorials, letters, and retracted items were removed by document type. Titles were then screened; for the three largest query sets (oil-palm imaging, depth and 3D sensing, and reviews of fruit detection and yield) titles were first sorted by keyword so that similar titles were read together. Abstracts were compiled from OpenAlex, Semantic Scholar, Crossref, and Europe PMC. For 254 records no abstract could be found at this stage; these were assessed from the title, the source, and, where available, the summary provided by Semantic Scholar.

Screening was carried out by one reviewer working with a large language model, which read the titles and abstracts and recorded a decision and a group for each record. To test the consistency of these decisions, the model was run a second time, in separate sessions that had no access to the first-pass decisions, on three sets of records: (i) all 510 records that the first pass had placed in the multi-observation group, placed in the oil-palm group, or excluded at the eligibility stage; (ii) a random sample of 300 records from the title stage; and (iii) a random sample of 100 records from the eligibility stage. For set (i) the second pass read the full text where one had been obtained, searched for an abstract where none had been found, and also recoded the multi-observation and oil-palm studies.

The two passes assigned the same group, or the second pass accepted the first-pass group as an alternative, for 85% of the 510 records. The 232 records on which any code differed were adjudicated from the source in a third run, and a proposed change of group was applied only if a further run that was instructed to refute it could not do so. This procedure changed the group of 16 records, left 7 proposed changes unapplied, and flagged 10 records whose group cannot be settled without the full text; unapplied and flagged records keep their first-pass decision. In the samples, the two passes agreed on 85% of the title decisions and on the group of 83% of the eligibility decisions.

Every disagreement at the title stage was a first-pass exclusion that the second pass would have carried forward (44 records); when their abstracts were read, 11 met an inclusion criterion, among them one multi-observation study and one oil-palm study. The abstracts of all 1,325 title-stage exclusions retrieved by the multi-observation and oil-palm query sets were therefore read in a further pass, and each proposed inclusion was again put to a run that tried to refute it; 19 records were included (6 multi-observation and 6 oil-palm studies among them), and they are counted throughout this article.

The first author also screened two random samples without seeing the model's decisions: 300 records at the title stage, and 165 records drawn from a random sample of 300 at the eligibility stage. The first author and the model agreed on 256 of the 300 title decisions (85%, Cohen's $\kappa = 0.57$) and on the full code of 86 of the 165 eligibility decisions (52%, $\kappa = 0.45$). The first author had marked 5 titles excluded by the model as possible multi-observation or oil-palm studies. The second author reviewed these and included 1, in the depth and 3D sensing group. The other 118 disagreements were resolved with the model. Two runs read each record from its source independently, a third run decided where they differed, and each resulting change was checked against the abstract or full text. This changed 8 decisions: 2 inclusions, 3 exclusions, and 3 changes of group.

Apart from the blind samples, these steps measure the consistency of a model-assisted procedure. They are not an independent review by a second person, and their consequences are discussed in Section 11. Every changed decision and code is recorded, with its old value, new value, and reason, in a change log provided as supplementary material.

Outside the search, 5 reports were identified. One review of citrus yield prediction, published online before the search date but assigned to a 2027 volume and therefore outside the year limit of the queries, was identified from a list of known papers and added to the earlier reviews (Liu and Ampatzidis, 2027). The reference lists of the five closest reviews were also checked for multi-observation studies that the search had missed. Of ten candidates, four met the criteria and were

Table 2
Groups of included studies (n = 1,018).

Group	Definition	Studies	How coded	Sections
Multi-observation studies	Combine several observations of the same fruit (video, several views, or repeated scans) and count, estimate, map, or link fruit across images	200	Individually, in two passes; from full text where available	3–6
Single-view counting studies	Count fruit or estimate yield from single images	245	Keyword rules, spot-checked	4, 5
Oil-palm studies	Image oil-palm fresh fruit bunches or loose fruit, including mill grading	171	Individually, in two passes	8
Single-image class-wise counting studies	Count fruit by class in single images	52	Keyword rules, spot-checked	6
Depth and 3D sensing studies	Detect, locate, or measure fruit with depth, 3D, or a non-RGB modality	139	Keyword rules, spot-checked	7
Earlier reviews	Review fruit detection, counting, yield estimation from images, or oil-palm imaging	123	Keyword rules; the closest reviews examined individually	9
Methods papers from outside agriculture	Tracking, re-identification, multi-view association, 3D reconstruction, or object counting	88	Keyword rules	3, 5

added: one multi-observation study (Apolo-Apolo et al., 2020a), two single-view counting studies (Payne et al., 2014; Kwon et al., 2024), and one depth-sensing study (Choi et al., 2015). These reports are counted in the group totals; the coded statistics and Appendix Table A1 cover the studies identified through Scopus.

2.5. Data extraction and coding

Each included study was assigned to one of seven groups (Table 2). Where a study met the definition of more than one group, the multi-observation group took precedence for any study that combines several observations of the same fruit. A stereo image pair was treated as a single observation: a study that detects, locates, or sizes fruit from one stereo pair was assigned to the depth and 3D sensing group, and to the multi-observation group only when it links fruit across frames or across camera positions. The map is broader than the four questions because each group has a role in answering them; Section 2.6 gives that role.

Each multi-observation study was coded for (i) association mechanism (one or more of the six defined in Section 3.3), (ii) acquisition design (video along a path, discrete views, 3D scan, revisits over time, or a single view), (iii) whether counts are reported per class, (iv) the main reported result, (v) the reference against which counts were checked, (vi) the level of the reported metric, and (vii), where counts are reported per class, how the class is attached to a unique fruit. Each oil-palm study was coded for (i) task, (ii) setting, (iii) modality, (iv) use of several views, and (v) whether class labels are attached to instances. The codes of the first pass were assigned from the abstract and, where full text was obtained, from the paper; the second pass described in Section 2.4 recoded both groups. The codes of all multi-observation and oil-palm studies were then assigned again by two runs that read each study without the earlier codes, under the rules below; a third run decided where the two differed, and the evidence quoted for each changed code was checked against the source. After this step, the codes of 195 of the 199 multi-observation studies identified through Scopus rest on the full text and 4 on the abstract.

The following rules decided the codes. A study that uses several mechanisms was assigned each of them, so mechanism totals are occurrences, not exclusive study counts. Linking between consecutive frames by motion or overlap was coded as temporal tracking. Trackers that add an appearance descriptor to the motion model, such as DeepSORT and StrongSORT, were coded as appearance matching combined with temporal tracking. Tracking followed by 3D localization was coded as temporal tracking combined with geometric association. A study that fuses images from several positions into a 3D model, panorama, or orthomosaic and detects or counts fruit in that model was coded as geometric association. A study that combines counts or features from two or more images of the same plant by a fitted or learned rule, without linking individual fruit, was coded as statistical correction; plain summation was coded as no association. A calibration applied to counts that another mechanism has already made unique was not coded as statistical correction, because it does not decide identity. A study that tracks fruit across frames was included as a multi-observation study only when the tracking or a count was evaluated. Acquisition was coded as a 3D scan when fruit were detected or counted in a point cloud or 3D model, and as revisits over time only when the identity of a fruit was linked across visits. Counts per class were coded only when a study reports counts, not only detections, for each class. Dataset papers without an association method were coded separately and are not included in the mechanism statistics.

The remaining groups were coded by keyword rules for crop, modality, platform, and class attribute, and the rule output was spot-checked. Metric types were extracted from abstracts by keyword and are reported only for studies with an abstract. Full text was obtained for 476 of the 1,013 included studies, among them 197 of the 199 multi-observation studies identified through Scopus; the full text of the other 2 could not be obtained, and their codes rest on the abstract. The codes of all 199 multi-observation studies

Table 3
Questions of the review, the groups of included studies that answer them, and the sections where they are answered.

Question or purpose	Main evidence	Supporting groups	Sections
(1) Mechanisms and the conditions each assumes	Multi-observation studies	Depth and 3D sensing studies; methods papers from outside agriculture	3–5, 7
(2) How well each mechanism works, and how that was measured	Multi-observation studies	Single-view counting studies, as a baseline	4, 5
(3) Classes on unique fruit	Multi-observation studies that report counts per class	Single-image class-wise counting studies	6
(4) The oil-palm case	Oil-palm studies	None	8
Positioning	Earlier reviews	None	9

identified through Scopus are listed in Appendix Table A1 and underpin the counts in Section 3.4 and the synthesis in Sections 4 to 8. The unit of these counts is the report: a conference paper and a journal article by the same authors on the same apple-counting pipeline (Osman et al., 2021b,a), and two reports on the same apple orchard datasets (Roy et al., 2019; Dong et al., 2020), are counted separately.

2.6. Analytical framework

The review is organized by a five-stage design space and by six association mechanisms, both defined in Section 3. Table 3 links each question to the groups that carry its main evidence, the groups that support it, and the sections that answer it.

2.7. Evidence appraisal and synthesis

No formal quality score was assigned. Instead, the review records, for each comparison it reports, the reference against which counts were checked (hand count on the tree, harvest count, packhouse count, or annotated images), whether the comparison was made on the same trees or data, and whether the result is a total count, a per-tree count, or an identity metric. These three properties decide what a reported number can support.

The synthesis used three procedures. First, descriptive statistics summarize the coded variables; statistics on mechanisms refer to the multi-observation studies that propose or evaluate a method, and statistics on metric types refer to the studies with an available abstract. Second, within each mechanism a structured narrative synthesis compares studies by acquisition design, crop, reference, and metric level; comparisons made on the same data are collected in Tables 4 and 5. Third, as a supplementary perspective, a term co-occurrence map was computed from the titles and available abstracts of the 1,013 included studies identified through Scopus (787 have an abstract; the others contribute their title only). A fixed vocabulary of task, method, sensor, and crop terms was matched in each record; terms found in at least 20 studies were kept, two terms were linked when at least 12 studies contained both, and terms were grouped by modularity clustering on association strength. The map was used for orientation only and did not affect inclusion.

2.8. Search results and included studies

The flow diagram (Fig. 1) summarizes the record counts at each stage. The searches returned 6,491 records, of which 5,889 were unique. After 166 were removed by document type, 5,723 titles were screened and 4,578 excluded. Of the 1,145 records assessed for eligibility, 132 were excluded, most often under exclusion criterion (v) (60 records). The map contains 1,018 studies: 1,013 identified through Scopus and 5 identified by another method (Section 2.4). Fig. 2 shows the growth of the Scopus records, from 9 in 2012 to 196 in 2025; multi-observation studies grew from at most 5 per year up to 2019 to 48 in 2025. Records for 2026 cover the period up to the search date.

Fig. 3 shows the term map: 43 terms and 317 links. Its three large clusters match the groups of the review: counting, yield estimation, video, and tracking sit together with the terms for identity (data association, re-identification, double counting, multi-view); depth, RGB-D, stereo, LiDAR, and point clouds sit with apple, orchard, occlusion, and localization; and ripeness, classification, and grading sit with oil palm. The identity terms are small and peripheral: each occurs in 24 to 37 of the 1,013 studies, against 621 for detection and 431 for counting.

3. A design space for turning observations into records

This section (i) describes the five stages that lead from images to a class-wise inventory, (ii) names the ways in which a fruit is counted twice, (iii) defines the six association mechanisms by the condition each assumes, and (iv) reports how the 199 multi-observation studies identified through Scopus are distributed over mechanisms, acquisition designs, crops, platforms, and sensors.

3.1. From images to an inventory

Fig. 4 shows the five stages of the pipeline that ends in a class-wise inventory. Acquisition fixes how many views of each plant exist, whether they are ordered in time, and whether pose, depth, or baseline are known. Observation produces detections or masks in each image; a fruit may produce several observations, and a detector may also split one fruit or merge two. Association decides which observations belong to the same physical fruit. Attribute assignment gives the unique fruit a class, size, or mass. The inventory

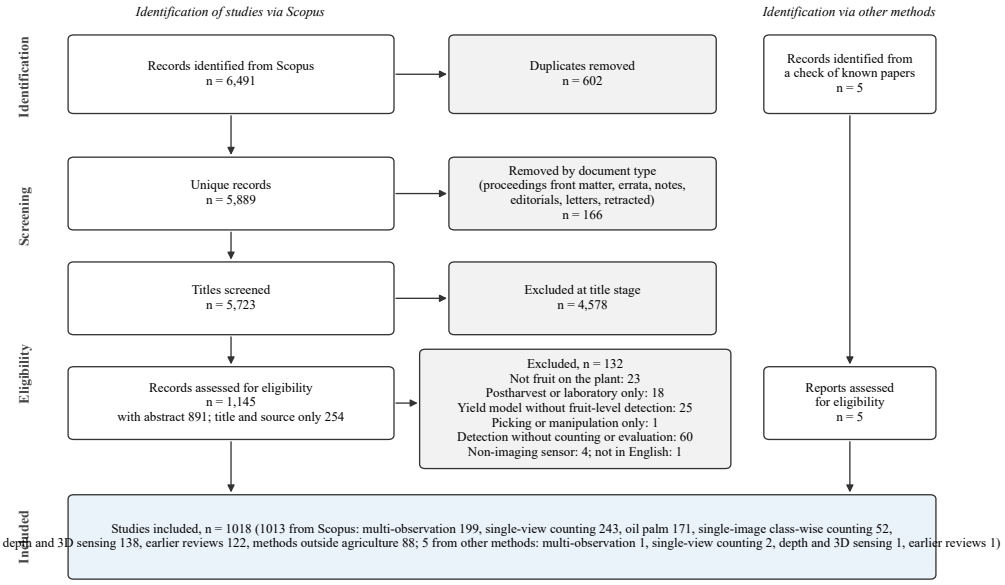


Fig. 1. PRISMA 2020 flow diagram of record identification, screening, eligibility assessment, and inclusion.

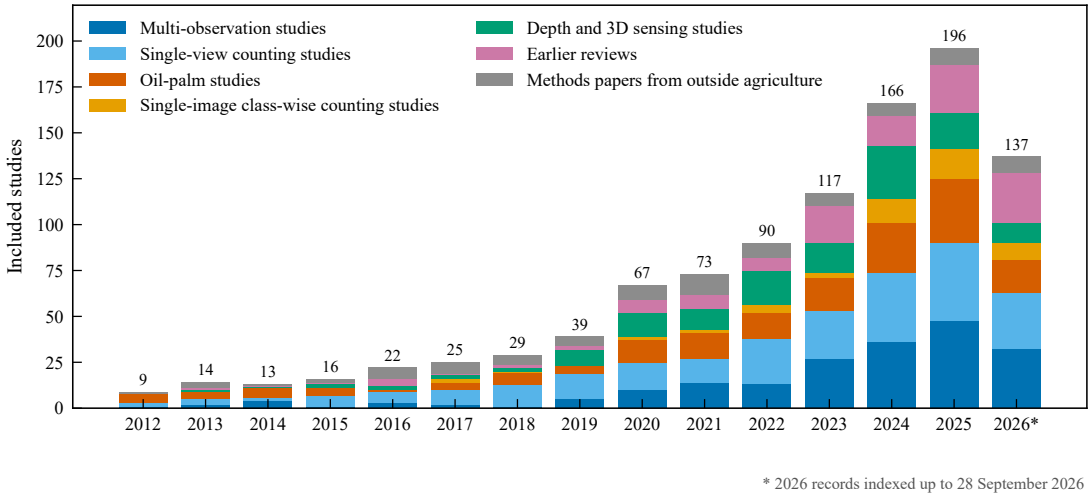


Fig. 2. Number of included studies per year by group (n = 1,013 studies identified through Scopus).

aggregates unique fruit by class and by tree, row, or block. Detection accuracy, which most papers report, describes the second stage. The count, which the user needs, depends on all five.

3.2. Three sources of double counting

Liu et al. (2019) named three ways in which a fruit is counted twice in image sequences of a row: the same fruit detected in consecutive images, the same fruit tracked, lost, and picked up again as a new track, and the same fruit seen from the two sides of a tree. Wang et al. (2019) took the same three sources as the starting point for their mango tracker. Misses have two sources: fruit never visible from any camera position, and fruit visible but not detected. A mechanism controls some of these sources and not others, which is why the mechanisms below are described by what they assume, not only by how they compute.

3.3. Six association mechanisms and the conditions they assume

No association. Each image is counted separately. This is correct only if acquisition guarantees that no fruit appears twice, or if duplicates are accepted.

Statistical correction. Image counts are summed or averaged and then related to hand or harvest counts by a fitted factor or regression. No individual fruit is identified. It assumes that the ratio between observed and true counts is stable across the trees to which it is applied (Nuske et al., 2014a; Linker et al., 2015).

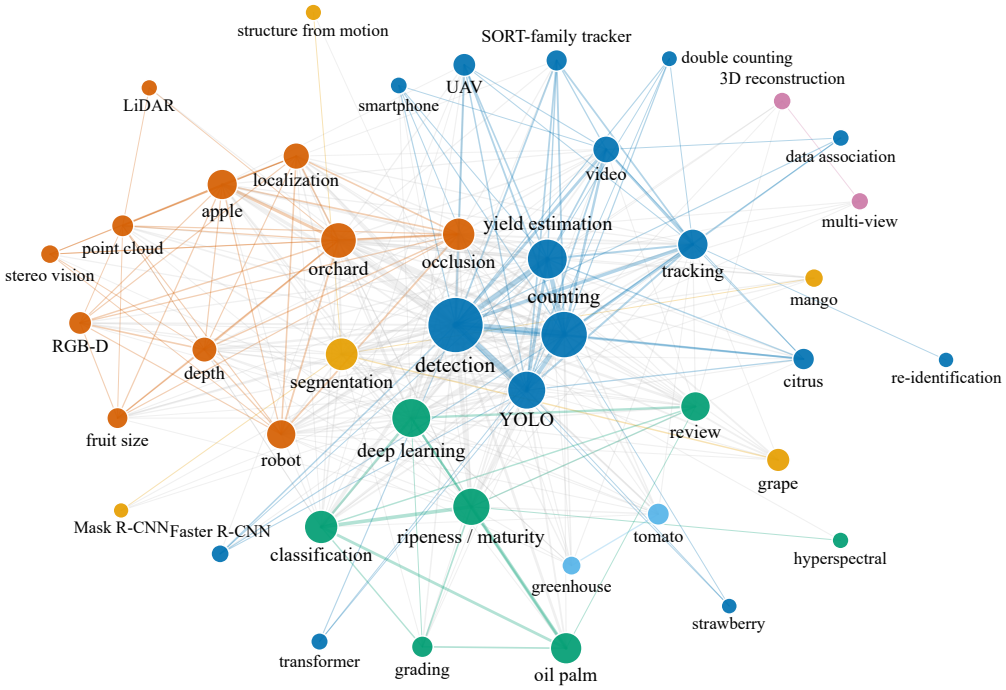


Fig. 3. Term co-occurrence map of the included studies, where node size, line width, and colour represent the number of studies containing the term in the title or abstract, the number of studies containing both terms, and the cluster, respectively.

Appearance matching. Observations are linked when their appearance descriptors are similar, either hand-crafted or learned for re-identification. It assumes that fruit differ enough in appearance to be told apart.

Temporal tracking. Observations in consecutive frames are linked by predicted motion and overlap, usually with a Kalman filter and an assignment step (Bewley et al., 2016); a counting line or region then decides when a track is counted. It assumes that consecutive frames overlap and that motion between them is small and smooth.

Geometric or 3D association. Observations are linked through a common coordinate frame: triangulation with known camera poses, structure from motion (SfM), simultaneous localization and mapping (SLAM), registered point clouds, or an implicit 3D representation such as a neural radiance field. It assumes that poses, scale, and depth or reconstruction are accurate enough that two different fruit are not placed at the same position.

Learned association. A network is trained to decide correspondence directly from linked examples. It assumes that training data with identity links exist and represent the conditions of use.

A study may combine mechanisms; 84 of the 195 multi-observation method studies do (Appendix Table A1). In tables and figures a plus sign joins the mechanisms that a study combines, for example appearance + tracking. A study that combines them should still show which one removes the duplicates, because each fails under different conditions.

3.4. Quantitative landscape of the multi-observation studies

To complement the synthesis in Sections 4 to 8, the coded records (Section 2.5) were summarized with descriptive statistics. Of the 199 multi-observation studies identified through Scopus, 4 are dataset papers and 195 propose or evaluate a method; the statistics below refer to these 195.

(i) Mechanisms and their change over time. Fig. 5(a) shows the share of method studies that use each mechanism in each period; because a study may combine mechanisms, the values are occurrences rather than mutually exclusive study counts. Temporal tracking appears in 126 of the 195 studies (65%), geometric association in 79 (41%), appearance matching in 57 (29%), statistical correction in 13 (7%), learned association in 8, and no association in 1. The mix has shifted. Of the 13 studies published before 2018, 10 used geometric association, 2 statistical correction, and 3 temporal tracking; since 2023, temporal tracking has appeared in 70% of the 141 new studies and appearance matching, mostly inside trackers that carry an appearance descriptor, in 35%.

(ii) Acquisition design. Fig. 5(b) shows that most studies acquire video along a path (138); 20 use discrete views, 33 use 3D scans, and 4 revisit the same plants over time. The literature is therefore mainly a literature of video.

(iii) Crops and class reporting. Fig. 6 relates crop to mechanism. Apple accounts for 52 studies, citrus for 21, grape for 21, tomato for 19, strawberry for 11, and mango for 7. Only 25 of the 195 studies report counts per class, which Section 6 takes up.

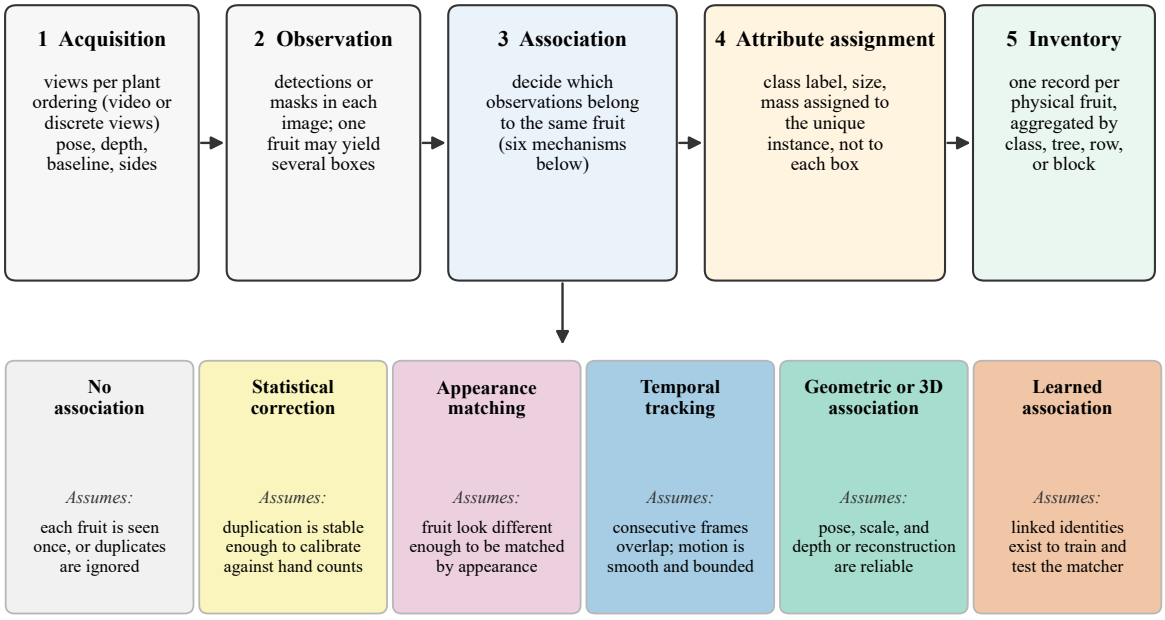


Fig. 4. Design space: five stages from acquisition to inventory, and the six association mechanisms with the condition each assumes.

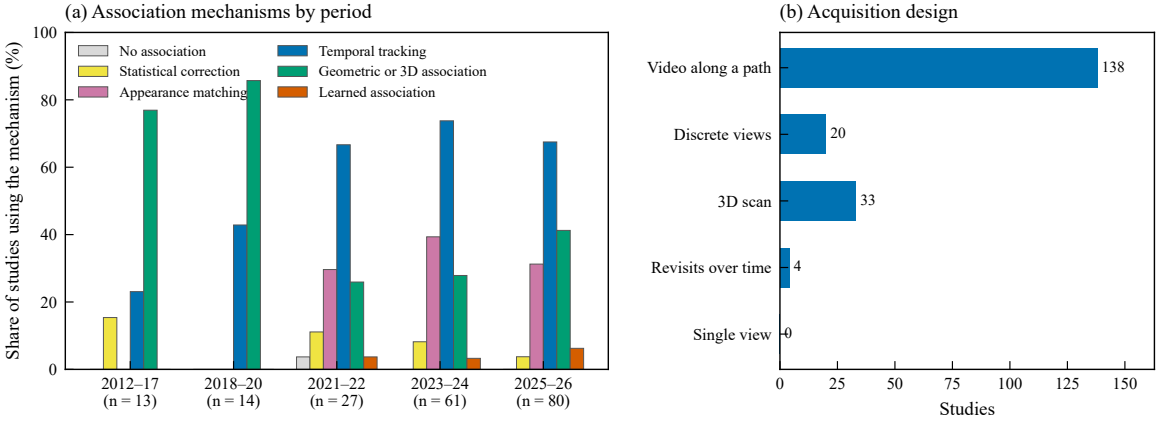


Fig. 5. Association mechanisms and acquisition design in the multi-observation method studies ($n = 195$): (a) share of studies using each mechanism by period; (b) acquisition design.

(iv) Combinations, from acquisition to crop. Fig. 7 follows each of the 195 studies from acquisition to crop. Here a study is placed in one category, so combinations are separate: temporal tracking alone accounts for 47 studies, geometric association alone for 52, appearance matching combined with temporal tracking for 52, and temporal tracking combined with geometric association for 19. Video feeds all four of these, whereas discrete views lead mainly to geometric association (12 of the 20 studies) and to statistical correction (8). Taken together, Fig. 5(a) counts every mechanism a study uses, whereas Fig. 7 places each study in one category; the first shows prevalence and the second shows which combinations occur.

(v) Platforms and sensors. Fig. 8 gives modality and platform by crop. The platform is the least reported property: 118 of the 195 abstracts do not state it, 39 name a ground vehicle or robot, 11 an unmanned aerial vehicle (UAV), 6 a handheld device or smartphone, and 17 name more than one platform; in Fig. 8(b) a study that names several platforms is counted under each. These counts describe what abstracts report, not what was used. An RGB camera is the sensor in 162 of the 195 studies; RGB-D cameras appear in 14, stereo cameras in 6, LiDAR in 7, and multispectral cameras in 4. Sensors that measure depth are concentrated in apple and strawberry.

Sections 4 to 8 use this design space. Section 4 takes the acquisition stage, Section 5 the association stage one

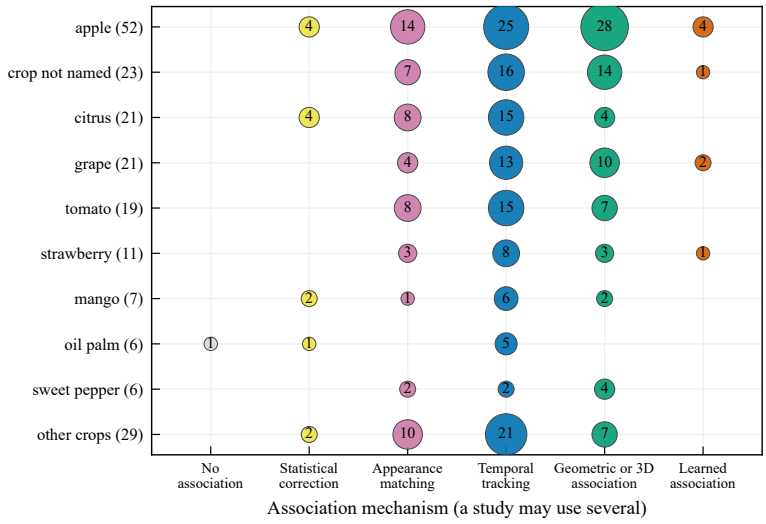


Fig. 6. Multi-observation method studies by crop and association mechanism (n = 195), where the number in each circle is the study count.

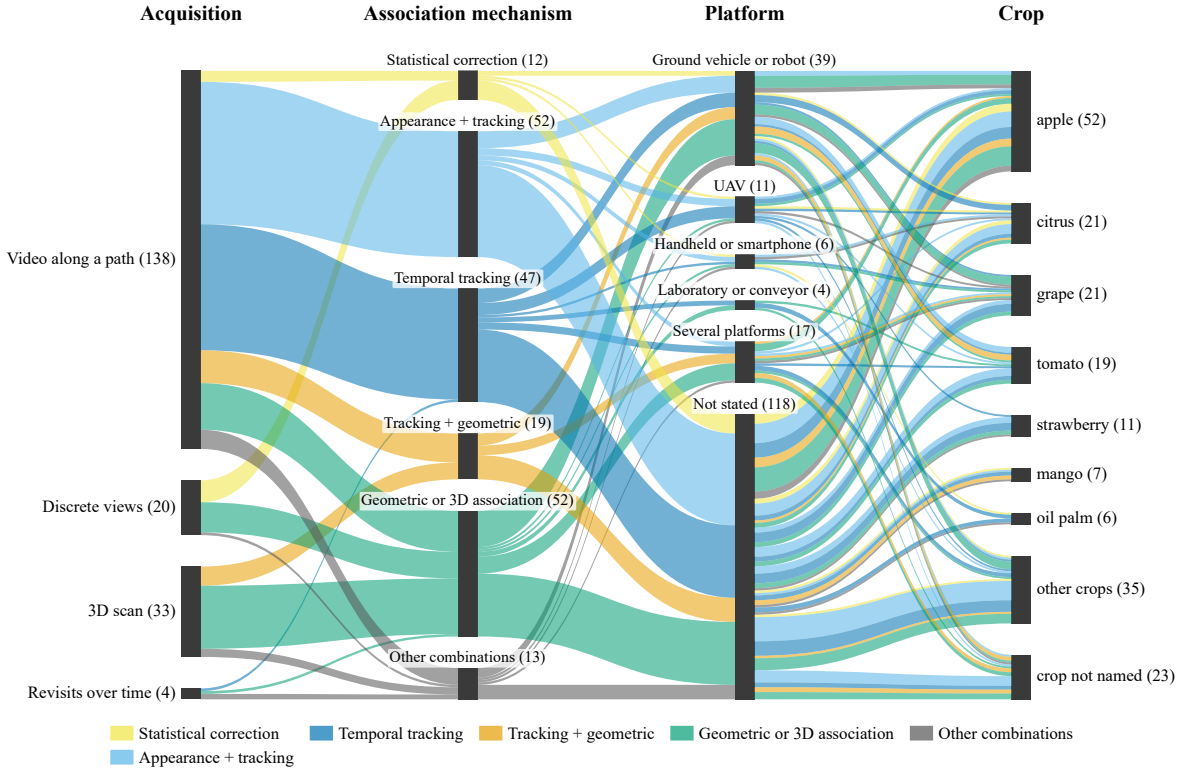


Fig. 7. Acquisition design, association mechanism, platform, and crop of the multi-observation method studies (n = 195), where band width and colour represent study count and mechanism, respectively.

mechanism at a time, and Section 6 the attribute stage; Sections 7 and 8 then ask what depth sensing and the oil-palm case add.

4. Acquisition: how many views, and from where

This section covers the acquisition stage. Eight studies compared acquisition designs on the same plants (Table 4); their results show how much recall additional views add and when they do not help. Each row is one study; the

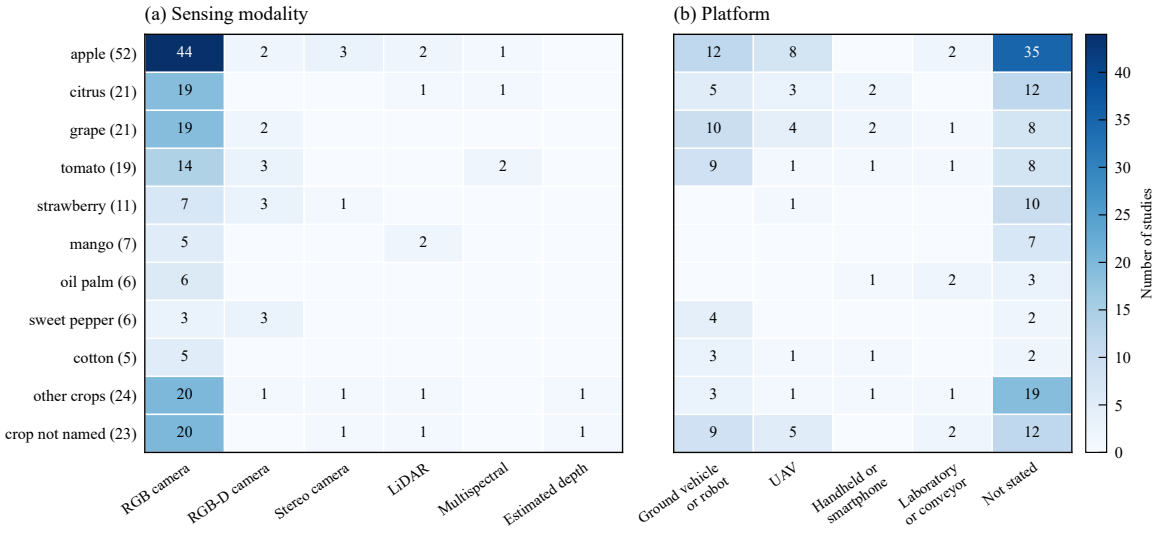


Fig. 8. Sensing modality and platform by crop in the multi-observation method studies ($n = 195$): (a) modality; (b) platform.

Table 4
Studies that compared acquisition designs on the same plants ($n = 8$).

Study	Crop, scale	Designs compared	Reported result	Count reference
Stein et al. (2016)	Mango, 16 trees	One image of one side; one image per side; multi-view tracking with triangulation (tracking + geometric)	Fruit counted: 27% (single, $R^2 = 0.81$), 54% (dual, $R^2 = 0.94$), 101.4% (multi-view, $R^2 = 0.90$)	Harvest count
Wang et al. (2019)	Mango, 21 trees	One image per side; video tracking	40% of harvest (bias-corrected RMSE 21.7 fruit/tree) vs 62% (RMSE 18.0)	Harvest count
Gongal et al. (2016)	Apple, tall spindle	One side; both sides with 3D duplicate removal (geometric association)	Crop-load estimation accuracy 58% vs 82%; error in identifying duplicate apples 21.1%	Manual count
Qureshi et al. (2023)	Apple fruitlets	One-sided vs two-sided scans from an over-canopy robot	Counting accuracy 73.7% vs 81.17%	Manual count
Ariza-Sentís et al. (2025)	Grape, commercial vineyard	Single-view vs multiple-view UAV video with tracking	Tracked/ground-truth ratio 13–23% vs 27–74% across trials; multi-view tracking degraded by motion blur	Annotated video
Felip-Pomés et al. (2026)	Apple, 420 trees	Two RGB-D sensors, three distances, two dates, one or both sides	Best MAPE 6.91%, $R^2 = 0.733$ at 225 cm; both sides reduced RMSE per stretch	Harvest count per 5 m stretch
Vijayakumar et al. (2023)	Citrus, per tree	UAV features plus fruit counts from one or two sides	MAPE 23.45% (one side) vs 25.72% (two sides), both better than UAV only (35.59%)	Harvest per tree
Anderson et al. (2021a)	Mango, commercial orchards	Raw multi-view counts vs multi-view corrected with hand counts	Absolute percentage error on packhouse count 11% vs 17% (nine orchards with all methods, 2019–2020 season); trellised canopies overestimated by 25.2% against harvest counts	Packhouse count; harvest count (canopy trial)

Note: Values are as reported by the authors. Count references and metrics differ between rows, so values should be compared only within a row. MAPE, mean absolute percentage error; RMSE, root-mean-square error; UAV, unmanned aerial vehicle.

values are as reported and should be compared only within a row, because references and metrics differ between studies (Section 5.6).

Three results are consistent across the eight comparisons.

(1) Single images per side count about half of the fruit or fewer. Stein et al. (2016) counted 27% of the fruit from one side and 54% from two sides of mango trees, and Wang et al. (2019) counted 40% of the harvest with one image per side. In apple, Gongal et al. (2016) reached a crop-load estimation accuracy of 58% with single-side imaging against 82% with dual-side imaging.

(2) Low recall does not prevent a precise estimate if it is calibrated. In the study of Stein et al. (2016), the dual-view counts correlated with the harvest counts more tightly ($R^2 =$

0.94) than the multi-view counts ($R^2 = 0.90$), but the dual-view counts had to be divided by a slope of 0.54, whereas the multi-view slope was 1.014. Precision and accuracy therefore answer different questions; a correlation with hand counts does not show that duplicates were removed.

(3) Adding views does not always help. Vijayakumar et al. (2023) found that counts from the second side did not reduce the per-tree yield error in citrus, whereas Ariza-Sentís et al. (2025) recovered more bunches with multiple-view flights over a vineyard but reported that motion blur made tracking harder in those flights. Zhu et al. (2024) found that the best number of views per citrus tree (two, four, or six) depended on how heavily the tree had been pruned. More views therefore raise recall but do not guarantee a lower error per tree.

How acquisition constrains association. Acquisition also decides which association mechanisms are possible. Video along a row gives the frame-to-frame overlap that temporal tracking needs. Discrete views around a tree, as in SawitMVC (Indriani et al., 2026) or the eight direction frames that Nayak Nenavath and Perumal (2024) sampled at equal intervals from a 360-degree walk-around video of a mango tree, have wide baselines and no frame-to-frame continuity, so tracking does not apply and association must rely on appearance, geometry, or learning. Several groups have treated view selection itself as the problem: active view planning for apple clusters (Roy and Isler, 2017a), viewpoint planning with an octree of fruit regions (Zaenker et al., 2021), and graph-based view sequences for sweet pepper (Zaenker et al., 2023). These studies optimize coverage or information gain; where they check fruit estimates against a reference, they do so in simulation or on artificial trees.

Taken together, the eight comparisons show that acquisition limits how much of the fruit can be seen and decides which association mechanisms can be applied; Section 5 examines those mechanisms.

5. Evidence by association mechanism

This section addresses the first two questions for the association stage: which mechanism was used, and what evidence shows that it controls duplicates and misses. Temporal tracking appears in 126 of the 195 method studies and geometric association in 79 (Section 3.4), so these two subsections are the longest. The subsections run from the mechanisms that identify no individual fruit, to linking in the image by motion and by appearance, to linking in a common 3D frame, and to correspondence learned from linked examples. Table 5 collects the 14 comparisons that were made on the same data; the values are as reported by the authors and can be compared within a row but not between rows. All studies coded for each mechanism are listed in Appendix Table A1.

5.1. No association and statistical correction

Statistical correction is used in 13 of the 195 method studies (7%); no association occurs in 1, and only alongside another mechanism (Section 3.4). Some studies limit duplicates mainly through acquisition. Nayak Nenavath and Perumal (2024) combined the mangoes detected in eight direction frames of a tree and subtracted those at the frame corners, which appear in two neighbouring frames; they counted 133 fruit against a manual count of 130 on one tree, with a counting accuracy of 95.48% on 14 trees. Turečková et al. (2022) avoided tracking by stitching a narrow vertical strip of every video frame into one panorama of a tomato row and counting once in that image, which the coding treats as geometric association; the net detected count differed from the annotated count by 2.32%, with false positives and misses partly offsetting each other. Villacrés and Vougioukas (2024) projected fruit from a reference camera into the other cameras of a calibrated four-camera rig and reported that one camera perpendicular to an apple canopy detected up to 88.3% of the fruit detected by four cameras.

The earliest multi-image systems calibrated their counts against harvest or hand counts, whichever step removed the overlap between images. Nuske et al. (2014a) registered images from a vehicle driving along vineyard rows and related berry counts to harvest yield; with calibration data from a prior harvest the estimate of total yield was within 4%, and within 3% with destructive samples taken at the time of imaging. Across four seasons their estimates captured up to 75% of the spatial variance in yield (Nuske et al., 2014b). Linker et al. (2015) photographed 42 apple trees from two sides at three heights, summed the six counts per tree, and calibrated the sum against 14 trees; the overall estimate for the 42 trees was within 10% of the actual yield, although single-tree errors were sometimes large. Song et al. (2014) mapped repeated, incomplete observations of pepper fruit in over 28,000 images of more than 1,000 plants to common fruit coordinates, corrected for the shift of each fruit between images, and grouped them with a statistical clustering step and reached a correlation of 74.2% with manual counts without linear adjustment.

The strength of statistical correction is repeatability. Stein et al. (2016) found the dual-view counts more tightly correlated with true counts than any other design, and Anderson et al. (2021a) reported absolute errors of 2% or less in repeated night-time imaging of mango orchards. The weakness is transfer. The factor describes an average tree and cannot say which fruit was counted twice. It must be refitted when canopy, cultivar, or acquisition changes. Anderson et al. (2021a) found the error on trellised canopies (25.2%) to be several times that on hedge, single-leader, or conventional systems (3.4–8.2%), and in the 2019–2020 season, across the nine orchards with all methods, correcting multi-view counts with hand counts of a stratified sample increased the average error from 11% to 17%. Regression from partial views to total counts remains common in UAV work: Apolo-Apolo et al. (2020b) related top-view apple counts to total counts per tree with $R^2 = 0.80$, and Li et al. (2023) used a regression across three views to account for the overlap between them.

Collectively, these studies show that statistical correction gives repeatable block-level estimates but no record of which

Table 5
Comparisons of association mechanisms on the same data ($n = 14$).

Study	Crop	Compared	Reported result	Metric level
Zhang et al. (2022a)	Citrus	OrangeSort (motion, counting region) vs SORT vs DeepSORT	Counting MAE 0.081 vs 0.45 vs 1.212	Count
Gené-Mola et al. (2023)	Apple	ByteTrack vs SORT vs DeepSORT (YOLOv5x)	HOTA 0.689, 0.680, 0.682; DeepSORT 128 ms per image vs about 15 ms; MAPE 7.47% one week before harvest	Identity and count
Piazolo et al. (2026)	Grape (UAV)	SORT vs SORT+ with extended matching cascade	Counting accuracy 33% to 96%; ID switches −62%; MOTA and IDF1 +5–6%	Identity and count
Ponce-Machete and Fajardo (2025)	Calamansi (UAV)	SORT, DeepSORT, ByteTrack	ByteTrack best: MOTA 59.67%, HOTA 65.42%, IDF1 73.72%	Identity
Wei et al. (2025)	Apple (stereo)	OC-SORT vs ByteTrack with GPS and visual odometry	OC-SORT best (HOTA 0.59); total error 20 apples, RMSE 6.24%, $R^2 = 0.96$	Identity and count
Qi et al. (2025)	Tomato clusters	SORT, DeepSORT, ByteTrack, BoT-SORT at reduced frame rates	Public MOT dataset of 15 videos with 1,614 tomato clusters	Identity
Liu et al. (2018)	Orange, apple	Tracking alone vs tracking plus SfM correction (tracking vs tracking + geometric)	SfM step rejected double-counted tracks and outliers; L1 count error fell from 593 to 203 (orange) and from 673 to 322 (apple)	Count
Häni et al. (2020)	Apple, three datasets	Summed counts of the two sides vs counts merged in a 3D reconstruction of both sides, with a ResNet50 counting network (no association vs geometric)	103.86–147.81% vs 95.56–97.83% of the harvested count	Count
Gené-Mola et al. (2020c)	Apple, 11 trees	2D detection vs detection projected into SfM (geometric)	F1 0.816 (2D, against visible fruit) and 0.881 (3D, test set of 1,021 apples on 8 trees)	Detection in 3D
Gené-Mola et al. (2020a)	Apple	LiDAR from one position vs multi-view, with and without forced air	6.5% (multi-view) and 6.7% (air flow) more fruit located	Detection in 3D
Lei et al. (2025)	Apple, 60 trees (association ground truth: five trees)	4D association (position, appearance, topology) vs position-only and histogram-descriptor baselines	Association precision +23.7% over the descriptor baseline; counting accuracy 96.9% (1,846 estimated vs 1,790 apples)	Identity and count
Hong et al. (2025)	Persimmon	3D Gaussian splatting vs NeRF-based counting	Counting accuracy 96%, reported as 16% above NeRF-based methods	Count
Xing et al. (2026)	Cherry tomato	Tracking with and without multi-view compensation	Counting MAE reduced by 95.6% against tracking without compensation	Count
Mollineda et al. (2025)	Orange	Regression from single frames vs video segments	Multi-frame segments better, tested with nonparametric statistics	Count

Note: Values are as reported by the authors and should be compared only within a row. Metric level: Count, agreement of the final count with a reference; Identity, tracking metrics such as multiple-object tracking accuracy (MOTA), the identification F1 score (IDF1), higher-order tracking accuracy (HOTA), or identity (ID) switches; Detection in 3D, detection F1 after projection. Metrics are given in the scale used by each paper (fraction or per cent). MAE, mean absolute error; MAPE, mean absolute percentage error; RMSE, root-mean-square error; SfM, structure from motion; NeRF, neural radiance field.

fruit was counted twice; it is the baseline against which the other mechanisms should be tested (Section 10).

5.2. Temporal tracking

Temporal tracking is the most common mechanism: it appears in 126 of the 195 method studies (65%) and in 70% of those published since 2023 (Fig. 5a). Most studies follow tracking by detection: a detector finds fruit in each frame, a Kalman filter predicts where each track should appear next, and an assignment step, usually the Hungarian algorithm on overlap or distance, links detections to tracks (Bewley et al., 2016; Wojke et al., 2017; Zhang et al., 2022c). Four issues recur in these studies.

(1) The counting rule. A counting rule decides when a track becomes a fruit. Two rules dominate: count each unique track identity, or count a track when it crosses a line or region. Parico and Ahamed (2021) found the unique-identity rule more reliable for pears (F1 87.85%); the line rule missed some fruit that were detected because the detections flickered. Zhang et al. (2022a) counted citrus tracks only within a fixed counting region, and this variant gave the lowest error in Table 5.

(2) Motion alone versus motion with appearance. The main question within temporal tracking is whether appearance features help. In the two comparisons on the same apple and citrus data, DeepSORT, which adds a learned appearance descriptor to the motion model, did not outperform trackers that use motion alone (Table 5; Zhang et al., 2022a; Gené-Mola et al., 2023). In the citrus study the counting error of DeepSORT was higher than that of SORT (MAE 1.212 against 0.45) (Zhang et al., 2022a), and in the apple study DeepSORT needed 128 ms per image

against about 15 ms for SORT and ByteTrack (Gené-Mola et al., 2023). Saraceni et al. (2024) designed AgriSORT to use motion only, arguing that fruit are static targets of similar appearance, and released a table-grape benchmark with strong self-similarity. Piazolo et al. (2026) improved SORT with a matching cascade over Mahalanobis, intersection-over-union, and Euclidean distances and reached accuracy comparable to DeepSORT without identity training data. The same issue appears outside agriculture: DanceTrack was built because targets with uniform appearance and diverse motion defeat trackers that rely on appearance (Sun et al., 2022). Studies that reported good results with DeepSORT exist, for example banana bunches with RGB-D and three constraints (96.82% counting success) (Zhou et al., 2025), but they did not compare against a motion-only tracker on the same data.

(3) Repairs for broken tracks. The failure modes that tracking studies report are specific. Missed detections break tracks and create new identities; Zhang and Cao (2025) added trajectory prediction to bridge such gaps in blueberry video, and Zhou et al. (2024b) added a multiple-mapping step that keeps strawberry counts consistent when track numbers jump. Camera shake on a legged robot reduced tracking accuracy until camera-motion compensation added 2.9 points of MOTA (Zhang et al., 2024), and PineSORT added a motion-direction cost and feature-based camera compensation, the latter for the large displacements between frames of low-frame-rate UAV video (Xie-Li and Fallas-Moya, 2025). Wang et al. (2019) reported double counts (9.9%) and misses (7.3%) separately, not only their net effect (an over-count of 2.6%).

(4) What tracking does not resolve. A fruit seen from both sides of a row appears in two separate passes. Wang et al. (2019), following Liu et al. (2019), listed it as a separate source of double counts, and Liu et al. (2019) used the depth of each fruit relative to its tree trunk to drop fruit that belong to the far side.

Taken together, these studies show that tracking links fruit across consecutive frames but cannot, by itself, resolve the third source of double counting (Section 3.2); the studies that address it use geometric association (Section 5.4).

5.3. Appearance matching

Appearance matching occurs in 57 of the 195 method studies (29%), almost always together with another mechanism: 52 studies combine it with temporal tracking through trackers that carry an appearance descriptor, such as DeepSORT, StrongSORT, or BoT-SORT with its re-identification module (Section 2.5). Kirk et al. (2021) trained the appearance network of a DeepSORT-based tracker with an additional ripeness branch to count soft fruit through occlusions and reported a count error (MAPE) of 3% against 21% for a baseline tracker. Xia et al. (2022) matched pairs of fruit patches between sequence images with a trained matching network and a one-to-one assignment, and reached F1 0.816 for apples and 0.864 for oranges in identifying repeated fruit. Zheng et al. (2023a) combined image appearance and position in a learned distance (an improved FaceNet) and clustered strawberry fruit and flower detections from overlapping multi-view remote sensing images; the result recovered objects that the orthomosaic hid. The pattern across these studies is that appearance works when combined with motion or position and has not been shown to work as the only cue for look-alike fruit, which agrees with the tracker comparisons in Section 5.2 and with the person re-identification literature, where discriminative identity features are learned from benchmarks with many labelled identities (Zheng et al., 2015; Ye et al., 2022).

5.4. Geometric or 3D association

Geometric association is used in 79 of the 195 method studies (41%); they are listed in Appendix Table A1 and shown by crop in Fig. 6. It is the route with the longest history of calibration-free counts. Five streams can be distinguished.

(1) Triangulation with poses from navigation sensors or stereo. Wang et al. (2013) used a stereo rig at night on an autonomous vehicle and registered sequential images; the yield error was −3.2% for about 480 red-apple trees and, after calibration with a small hand-measured sample, 1.2% for about 670 green-apple trees. Stein et al. (2016) established pairwise correspondences between images from the trajectory of a navigation system, triangulated tracked fruit, and used LiDAR to assign each fruit to its tree; 522 trees and 71,609 mangoes were scanned, and the multi-view count of the 16 validation trees was within 1.36% of the harvest count.

(2) Structure from motion from a single camera. Liu et al. (2018) limited the sensor input to a monocular camera: fruit were tracked in 2D and localized in 3D by structure from motion, and double-counted tracks were detected by their proximity in 3D. The later version also tracked trunks, built semantic structure-from-motion landmarks, and projected them back into every image (Liu et al., 2019). Roy and Isler (2016) built a piecewise incremental SfM for apple rows because standard SfM failed under the constrained geometry of a single camera moving along a row.

(3) Merging the two sides of a row. Roy et al. (2019) merged counts from both sides of the row on three validation datasets (6, 10, and 6 trees) with an accuracy of 91.98–94.81% against harvested counts. Häni et al. (2020) reused this pipeline with a learned counting network and compared it with a plain sum of the two sides: the sum gave 103.86–147.81% of the harvested count, and the merged reconstruction 95.56–97.83% (Table 5). Dong et al. (2020) addressed the same two-side problem by aligning the reconstructions of both sides with global and semantic features, since the two sides share little overlap. Gongal et al. (2016) used 3D positions to remove apples seen from both sides and reported an error of 21.1% in identifying the duplicates.

(4) Counting in a 3D map. Newer geometric systems move the whole count into 3D. Gené-Mola et al. (2020c) projected Mask R-CNN detections into an SfM point cloud

and improved F1 from 0.816 to 0.881, at a processing cost they judged unsuitable for real time. Matos et al. (2024) combined SfM with bipartite graph matching for apples filmed with unconstrained camera motion and low frame rates. Lee et al. (2026) fused YOLO detections with SLAM point clouds from LiDAR in a 6 m by 70 m pear plot and reached 96.2% counting accuracy. Nellithimaru and Kantor (2019) built object-level SLAM from stereo for grape counting (F1 0.977 against image counts). Rapado-Rincón et al. (2023) tracked partial point clouds of tomatoes in 3D and estimated the total count with a best-case counting error of 5.06%; they also showed that the final count of a tracker can hide false positives and false negatives that offset each other, and they applied tracking metrics to expose them.

(5) Implicit representations. FruitNeRF trains a semantic neural radiance field from posed images and clusters the extracted fruit points (Meyer et al., 2024), and Gaussian splatting was reported to count persimmons with 96% accuracy, 16% above NeRF-based counting (Hong et al., 2025). These methods build on general reconstruction tools (Schonberger and Frahm, 2016; Mildenhall et al., 2020; Kerbl et al., 2023); their evaluations are still dominated by synthetic scenes and a few real trees.

Where geometry fails. Scale is undefined for monocular reconstruction unless another sensor fixes it. Pose drift, where it occurs, enlarges the region within which two observations are declared the same fruit, so dense clusters can merge; no included study quantified this effect. Foliage moves between views, which violates the static-scene assumption of standard SfM and SLAM; dynamic-scene methods exist (Bescos et al., 2021; Saputra et al., 2019), but no included fruit study measured how much foliage motion degrades counts. Assigning fruit to trees is a separate geometric problem; Zine-El-Abidine et al. (2021) addressed it in a feasibility study by aligning a harvest-time point cloud to a leaf-off winter cloud in which each tree's branches are visible (accuracy above 95%).

Overall, geometric association has produced counts close to the reference without a fitted factor when pose, scale, and depth are reliable; the evidence for implicit representations still rests on synthetic scenes and a few real trees.

5.5. Learned association

A learned component decides the correspondence in 8 of the 195 method studies (Section 3.4), in three ways. First, a network matches fruit between scans or images. Fusaro et al. (2026) segmented fruit in coloured point clouds (terrestrial laser scans of strawberries and photogrammetric reconstructions of apples), extracted a descriptor per fruit with a sparse convolutional network, and matched fruit between sessions with an attention-based network and a probabilistic assignment, the matching being evaluated on the strawberry data only. The matching network of Xia et al. (2022) and the learned distance of Zheng et al. (2023a), discussed under appearance matching (Section 5.3), belong to the same group. Second, a network learns identity inside a 3D representation. FruitNeRF++ lifts 2D instance masks that are inconsistent between views into 3D through instance embeddings trained with a contrastive loss; on a real apple set it reached F1 0.765, below the 0.881 of the fruit-specific method of Gené-Mola et al. (2020c) (Meyer et al., 2025). Zhao et al. (2026b) combined cross-view contrastive learning with voting in a 3D Gaussian representation and reported F1 above 0.97 for plum and mango scenes with dense occlusion. Third, a network replaces the association step of a tracker. Santos (2026) used a learned point tracker for data association in orange video; on seven sequences the count was 978 against 879 annotated fruit (an error of 11.26%), which did not improve on the authors' earlier pipeline. A network can also fuse the views without deciding correspondence: Isobe et al. (2025) fused features from different angles of a 360-degree mandarin video in a transformer and regressed the count, without an explicit identity for each fruit, which the coding treats as statistical correction.

Outside agriculture, learned association is established for tracking of people and vehicles, within a single camera on pedestrian benchmarks (Braso and Leal-Taixe, 2020) and across cameras, where datasets with identities across cameras exist (Ristani et al., 2016; Chavdarova et al., 2018; Tang et al., 2019). The obstacle in fruit is the same one these datasets removed: learned association needs identity labels across observations (frames, views, or visits), and in the included corpus only a handful of fruit datasets provide them, among them GrapeMOTS (Ariza-Sentís et al., 2024), the tomato MOT set of Qi et al. (2025), the 4D orchard set of Lei et al. (2025), and SawitMVC (Indriani et al., 2026). This limit is taken up in Section 10.

5.6. How the evidence was measured

Of the 149 multi-observation method studies with an abstract, 64 (43%) report agreement between estimated and reference counts, 58 (39%) report detection quality, 18 (12%) report an identity metric such as multiple-object tracking accuracy (MOTA), the identification F1 score (IDF1), higher-order tracking accuracy (HOTA), or identity (ID) switches, and 53 (36%) report no quantitative metric in the abstract (Fig. 9). Identity metrics were developed for this purpose: HOTA balances detection and association accuracy (Luiten et al., 2021), and IDF1 was designed for applications in which identities must be preserved, within and across cameras (Ristani et al., 2016). A count error alone cannot separate a correct count from offsetting duplicates and misses (Rapado-Rincón et al., 2023; Wang et al., 2019).

The reference also varies. Hand counts on the tree measure what the camera could have seen, harvest and packhouse counts include fruit hidden from every view, and annotated images measure only what an annotator saw. In the coded records, the count reference is annotated images or video in 125 of the 195 method studies, a hand count on the plant in 43, a harvest count in 28, and a packhouse count in 1; 19 state no count reference in the source that was read (a study may use more than one). Results from different

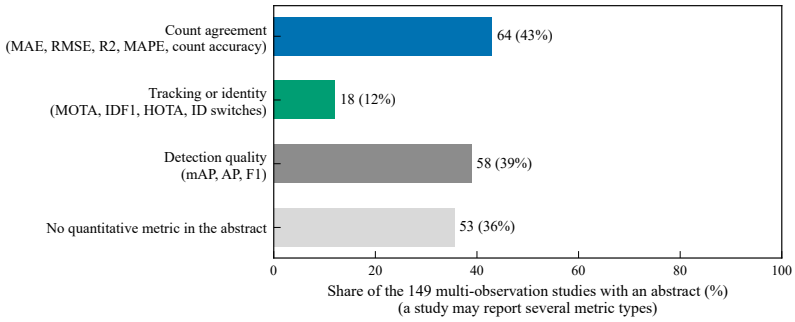


Fig. 9. Metric types reported in the abstracts of multi-observation method studies ($n = 149$).

references are not comparable, which is why Tables 4 and 5 report the reference or metric level for every row.

In answer to the first two questions, the included studies use six mechanisms whose assumptions are met by different acquisition designs (Section 4), and the evidence that a mechanism controls duplicates and misses rests mostly on count agreement: 14 comparisons were made on the same data (Table 5) and 18 of the 149 abstracts report an identity metric. Section 10 turns these observations into gaps.

6. Class attributes on unique fruit

This section covers the attribute-assignment stage and answers the third question: how class attributes, mainly maturity, are attached to a unique fruit rather than to each detection. Two bodies of evidence bear on this.

Classes counted in single images. The first is the 52 studies that detect and count fruit by class in single images. Most concern tomato (18) and blueberry (7), and 46 of the 52 use maturity as the class. MacEachern et al. (2023) detected green, red, and blue wild blueberries with YOLOv4 and predicted yield with nonlinear regression models, whereas Ni et al. (2020) counted blueberries with $R^2 = 0.886$ against ground truth. Deng et al. (2024) released 17,955 annotated ripe and unripe blueberries in canopy images, and Shimomoto et al. (2025) predicted sweet pepper yield from average counts at two ripeness stages. These studies count classes in the image, not classes of unique fruit.

Class assigned during tracking. The second is the 25 multi-observation studies that report counts per class. Where the source states how the class is attached, it is most often carried by the tracker (10 studies). Ogundele et al. (2025) used a class-aware ByteTrack for strawberries; counts correlated with ground truth at $R^2 = 0.914$ overall and at $R^2 = 0.950$ for ripe fruit. Zhou et al. (2024b) reported a relative counting error of 6.7% for all strawberries, and in a separate evaluation on two farms the lowest errors were 8.7% for ripe and 9.9% for unripe fruit. Wang et al. (2025a) counted tomatoes in a counting region using ByteTrack identities and obtained a mean counting error of 6.6% for all fruit but 12.3%, 16.9%, and 17.3% for the ripe, semi-ripe, and unripe classes. Zheng et al. (2025), who assigned the class after tracking, counted ripe tomatoes with an accuracy of 92.03–93.22% and unripe tomatoes with 89.74–92.55% against manual counts on three beds.

Class assigned by a vote. Disagreement between frames is resolved explicitly in 2 studies: Lin et al. (2026) tracked tomatoes with StrongSORT and gave each tomato the maturity class that received the most votes over the frames in which it appeared, and Qi et al. (2024) gave each tomato cluster the ripeness level that occurred most often across frames. Pai and Lin (2024) motivated a vote by inconsistent labels of the same tomato across frames, but voted among several models and evaluated each box without tracking. Pardo-Beainy et al. (2026) assigned each tracked strawberry to one of nine size–maturity categories from metric length and width derived from RGB-D point clouds and a colour-based ripeness percentage, and georeferenced the fruit with a global navigation satellite system for yield maps. Wu et al. (2025) reached an average counting accuracy of 72.49% across blue, purple, and green blueberries in video. Other studies classify tracked fruit but do not report counts per class: Hassan et al. (2025) tracked oranges with ByteTrack and classified tracked fruit as ripe or unripe with a separate network, and Viveros Escamilla et al. (2024) tracked sweet peppers of four colours with YOLOv5 and DeepSORT and reported 85.7% counting accuracy (MOTA 0.88 and 0.87) in two simulated environments, each a video assembled from 70 still greenhouse images.

Two consequences. Two properties of class-wise counting follow from these results. First, class errors and identity errors interact. A fruit seen in ten frames can receive different labels in different frames, especially near class boundaries, and a class-aware tracker that splits the track when the label changes creates a duplicate. In only 2 of the 25 studies is a rule stated for choosing the label when frames disagree, 12 do not state how the class is attached, and none of their abstracts reports a class confusion matrix computed on unique fruit. Second, per-class errors differ from the overall error. In the strawberry study of Ogundele et al. (2025) the ripe class had the highest count R^2 (0.950) and the lowest MAE (2.65 fruit), though not the lowest MAPE or RMSE (MAPE 11.63% against 9.62% for unripe), and in the tomato study of Wang et al. (2025a) each class error was 1.9 to 2.6 times the error for all fruit. In oil palm a video-based detector reached a mean average precision of 90.56% for a

single class but 70.21% for several ripeness classes (Junior and Suharjito, 2023). An inventory that reports only a total hides both effects (Section 10).

7. Depth and other modalities

Depth is examined separately because it can serve two stages of the design space: observation (finding, locating, and sizing a fruit) and association (placing observations in a common frame). The 139 depth and 3D sensing studies use RGB-D cameras (81), stereo (20), LiDAR (13), or other non-RGB signals. Their main use is not association but detection, localization, and sizing of individual fruit, most often apples (48). Four uses recur.

(1) Background removal. Fu et al. (2020b) filtered objects farther than 1.2 m from a Kinect V2 in a dense fruiting-wall orchard before detection.

(2) Fusion of colour and depth. Tu et al. (2020) trained separate colour and depth detectors for passion fruit and fused them late; the full method raised F1 from 0.885 for a standard Faster R-CNN on RGB-D images to 0.946. Kaukab et al. (2024) filled depth holes before fusion because outdoor depth images are usually of low quality.

(3) Metric size. Wang et al. (2017) estimated on-tree mango dimensions from RGB-D with root-mean-square errors of about 4–5 mm but noted that the camera could not be used in direct intense sunshine. Neupane et al. (2021) compared eight depth cameras against a 20 mm bias-corrected error requirement for fruit localization and sizing.

(4) Other active sensing. LiDAR detected apples by reflectance with F1 0.858 (Gené-Mola et al., 2019), and LiDAR–camera fusion reduced localization error to one fifth of that of a RealSense D455 at 1.8 m (Kang et al., 2022). Active thermal imaging with water mist separated immature citrus from leaves of similar colour and supported tracking-based counting in video (Gan et al., 2020).

Depth for association. For association, depth enters through geometric association (Section 5.4). Zheng et al. (2025) filtered detections by depth and used robot odometry in inter-frame prediction for tomatoes, reaching MOTA 0.954 against YOLOv8 with DeepSORT. Abeyrathna et al. (2023) obtained 3D coordinates of each tracked apple from a RealSense D455 at three vehicle speeds and three camera angles. Felip-Pomés et al. (2026) found that the Azure Kinect gave more consistent results than the ZED 2 across dates in a 420-tree orchard. Liu et al. (2019) used depth relative to the trunk to resolve two-side duplicates. Monocular depth networks now offer depth without a depth sensor (Yang et al., 2024a; Ranftl et al., 2021), and one included study used such a network with amodal masks to measure apple diameter (Tan et al., 2025a), but none used predicted depth for cross-view association with a counted reference. The evidence therefore supports depth as an aid to separating neighbouring fruit and to placing fruit in a common frame when its error is small relative to fruit spacing, and it identifies the conditions (sunlight, range, registration) under which that error grows. Section 8 returns to these conditions for oil palm.

8. Oil palm

This section answers the fourth question: what the oil-palm imaging literature covers, and what is missing for a bunch-level, class-wise census from images. It covers ripeness classification, detection, counting, and what the case implies for the choice of mechanism. Fig. 10 shows that the 171 oil-palm studies concentrate on ripeness: 76 classify the ripeness of images of single bunches, 42 detect bunches in a scene, 16 study non-RGB sensing of ripeness, 18 address quality, defects, mass, or pose, 12 detect loose fruit, 3 count bunches or fruit, and 4 are dataset papers. The setting is on the palm in 13 studies, at a mill, conveyor, or laboratory in 11, on the ground or at a collection point in 8, and from a UAV in 3; 136 sources do not state it. Class labels are attached to instances in 131 studies. Only 17 use more than one view of the same bunch or palm.

8.1. Ripeness classification and non-RGB sensing

Ripeness classification has a long record. Early work extracted colour features and classified them with neural networks or support vector machines (Fadilah et al., 2012; Sabri et al., 2018), followed by colour and texture features classified with a back-propagation neural network (Septiarini et al., 2021) and by convolutional networks deployed on mobile devices (Suharjito et al., 2021). One study used 10,000 photographs from five Indonesian plantation sites in four classes and reported 94.66% $\pm$ 0.54% accuracy in leave-site-out cross-validation on sites not seen in training (Usman et al., 2026). Reviews of this work agree that computer vision with deep learning is the most feasible field method (Lai et al., 2023) and that fusion with LiDAR may add localization (Goh et al., 2025a). Non-RGB sensing has been tested with thermal imaging (Zolfagharnassab et al., 2022), hyperspectral imaging (Bensaeed et al., 2014), laser-light backscattering (Mohd Ali et al., 2020), LiDAR (Husin et al., 2021), and radar with bunch mass (Resta et al., 2026), several of them on harvested bunches.

8.2. Detection on the palm and at the mill

Detection on the palm moved from single-class to multi-class ripeness detectors. Lai et al. (2022) trained YOLOv4 on images from a RealSense D435 with expert labels for ripe and unripe bunches, and Mansour et al. (2022) compared detectors on four ripeness classes. Junos et al. (2021) detected bunches, grabbers, and palms in plantation scenes, and the same group detected loose fruit in UAV images (Junos et al., 2022). Zaman et al. (2025) addressed occlusion by fronds and neighbouring bunches in UAV images. Goh et al. (2025b) released 400 plantation images with registered depth maps and point clouds from a RealSense D455f for ripeness classification and localization, the only released oil-palm RGB-D dataset in the corpus. At the mill, Suharjito et al. (2023a) graded piles of bunches from smartphone video and released annotated datasets of piles (Suharjito et al., 2023b). Detection in oil palm therefore has field data, class labels, and one depth dataset, but each image is evaluated on its own.

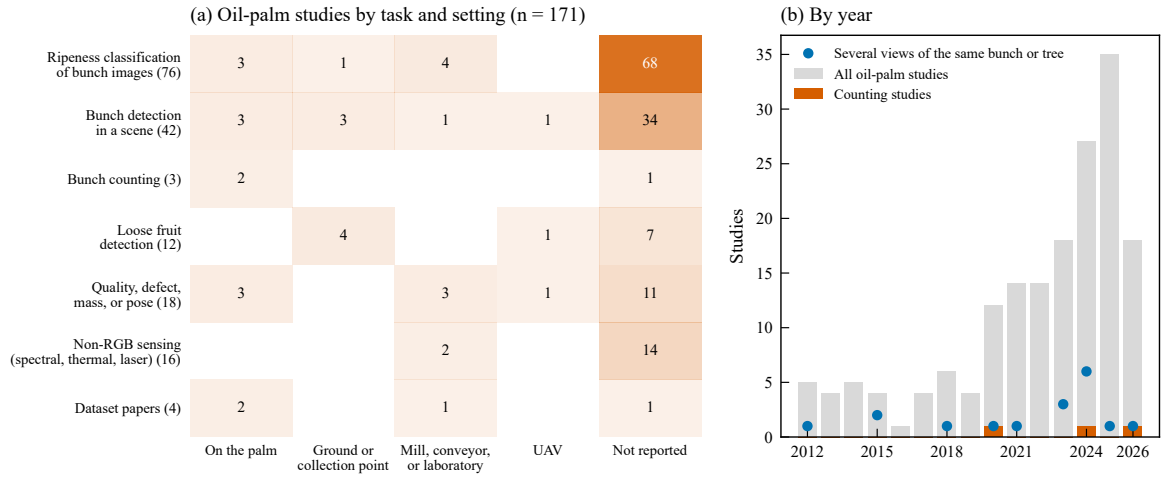


Fig. 10. Oil-palm studies ($n = 171$): (a) task by setting; (b) studies per year, with counting studies and studies that use several views marked.

8.3. Counting

Counting is the gap. Only 3 oil-palm studies evaluate counts as a main result, and each counts in single images: bunches in plantation video frames (Grady Naftali et al., 2024), bunches in images collected from the internet (Prasetyo et al., 2020), and fruit within bunches (Hamdani et al., 2026). Two further studies name counting in their aim or title but do not evaluate counts (Rizky Hidayat et al., 2024; Hutapea et al., 2024). Grady Naftali et al. (2024) built a plantation dataset of video frames with partial visibility, occlusion, small objects, low contrast, and blur, and Rizky Hidayat et al. (2024) proposed a standard procedure for photographing harvested bunches at the collection point.

Five oil-palm studies link observations across images and are therefore counted with the multi-observation studies. Two count bunches on a conveyor with a detector and a centroid tracker: fresh bunches in a laboratory (Shiddiq, 2023) and unstripped bunches in a mill (Aji et al., 2021). Two track the palm, not the bunch, across video frames and take the most frequent per-frame count of the loose fruit under it (Daud et al., 2022, 2023). One tagged 809 bunches and photographed them repeatedly during development, 217 of them from anthesis to harvest, linking the images through the tag (Kassim et al., 2012).

None of these studies associates bunches automatically across views of the same palm. Suharjito et al. (2025) recorded unharvested palms from multiple angles with a smartphone for a digital census and released 14,503 images in five maturity classes, but the boxes are not linked across images. The only oil-palm dataset in the corpus with cross-view identity links is SawitMVC, whose baseline experiments count bunches with a regression from per-view counts: placed 96.81% of the per-class counts within one bunch of the reference when it was given the annotated boxes and 75.35% when it was given detections (Indriani et al., 2026). Warni et al. (2026) combined stereo detection with Hungarian matching to localize loose fruit in 3D, which is an association step, but for a stereo pair, not for views around a palm. Counting from UAV images has addressed the palms themselves (Pragatheesh and Ramya, 2025; Kipli et al., 2023), not their bunches.

8.4. Implications for the choice of mechanism

The oil-palm case differs from the orchard cases in ways that matter for mechanism choice. Bunches of the same class on one palm look alike, which weakens appearance matching (Section 5.3). A census photographs a palm from a few discrete positions, which removes the frame-to-frame continuity that temporal tracking needs (Section 5.2). Fronds occlude bunches differently from each side, and the canopy is tall, which makes depth from consumer RGB-D cameras less reliable at the required range (Section 7). The census classes are defined by months to harvest, so neighbouring classes differ in subtle colour changes, and a class error on one view can split one bunch into two records of different classes (Section 6). These four properties define the missing experiment that Section 10 takes up.

9. Position relative to earlier reviews

With the evidence laid out, this section returns to the earlier reviews named in the Introduction and states what this review adds to each line of work. Table 6 lists the earlier reviews that are closest to this one. The column "Stated method" records what each review states about how its studies were found, from the abstract or, where it was read, from the full text. Three reviews are organized by sensor or sensing technique (Gongal et al., 2015; Fu et al., 2020a; Goh et al., 2025a), three by method, architecture, or model type (Koirala et al., 2019; Maheswari et al., 2021; Pu et al., 2025), three by estimation route, data source, or input feature (Anderson et al., 2021b; Rong et al., 2026; Liu and Ampatzidis, 2027), and two oil-palm reviews by task (Khan et al., 2021) or by sensing modality and classifier (Lai et al., 2023). Reviews of tracking and re-identification outside agriculture (Luo et al., 2021; Rakai et al., 2022;

Table 6
Earlier reviews closest to this review ($n = 11$).

Review	Scope	Organizing variable	Stated method
Gongal et al. (2015)	Fruit detection and localization for harvesting and crop-load estimation in apples, pears, citrus	Sensor and system	Not stated
Koirala et al. (2019)	Deep learning for fruit detection and yield estimation; extrapolation of image counts for occluded fruit	Method and application	Not stated
Fu et al. (2020a)	Consumer RGB-D cameras for fruit detection and localization	Depth principle and sensor	Not stated
Anderson et al. (2021b)	Forecasting fruit load and harvest timing	Estimation route (sampling, vision, remote sensing, models, harvest timing)	Not stated
Maheswari et al. (2021)	Semantic segmentation for fruit yield estimation	Workflow stage and network architecture	Not stated
Khan et al. (2021)	Machine learning in oil palm, 2011–2020	Task (regression, classification)	Systematic, 61 papers
Lai et al. (2023)	Oil-palm fresh-fruit-bunch ripeness detection	Sensing modality and classifier	Systematic, 51 papers
Goh et al. (2025a)	Fresh-fruit-bunch ripeness classification	Sensing technique	Self-described systematic; no databases or search strings reported
Pu et al. (2025)	UAV counting of field crops	Agronomic target and model type	52 key studies
Rong et al. (2026)	Orchard yield estimation with multi-source sensing	Data source	PRISMA
Liu and Ampatzidis (2027)	Citrus yield prediction: input features, acquisition platforms and feature extraction, prediction models, and related estimation, monitoring, and measurement methods	Input feature, acquisition platform, and prediction model	Self-described systematic; search details not given in the abstract
This review	Turning repeated fruit observations into unique, class-labelled records; oil palm as the main case	Association mechanism and its assumptions	Systematic map, Scopus, 1,018 studies

Note: None of the earlier reviews uses identity across observations as its organizing variable. Not stated: the abstract does not state how studies were found. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; UAV, unmanned aerial vehicle.

Ye et al., 2022) and of 3D reconstruction (Saputra et al., 2019; Zheng et al., 2026) provide the mechanisms but not the agricultural evidence.

None of these reviews organizes the evidence by identity across observations, although several discuss occlusion and extrapolation from partial counts; the gaps that follow from doing so are set out in Section 10.

10. Gaps and a measurement agenda

This section synthesizes Sections 4 to 8. Rather than repeating individual studies, it first traces how the choice of mechanism has shifted, then answers the four questions, and then states five gaps, each tied to the measurement that would close it.

10.1. How the choice of mechanism has shifted

Before 2018, geometric association and statistical correction accounted for almost all multi-observation studies; since 2023, temporal tracking has appeared in 70% of new studies (Fig. 5a). Within geometric association, implicit 3D representations are the latest variant, and 8 studies learn part of the association (Sections 5.4 and 5.5). The trajectory therefore runs from correcting a sum, to linking detections in video, to placing fruit in a 3D frame. Class-wise identity has not followed: only 25 of the 195 method studies report counts per class (Section 6). The term map agrees: each identity term occurs in 24 to 37 of the 1,013 studies (Fig. 3).

10.2. Answers to the four questions

Which mechanisms have been used, and what does each assume? Six mechanisms are in use (Section 3.3). Temporal tracking assumes overlapping, ordered frames and is therefore tied to video; geometric association assumes reliable pose, scale, and depth; appearance matching assumes fruit that can be told apart; statistical correction assumes a stable ratio between observed and true counts; and learned association assumes identity-labelled data (Sections 4 and 5).

How well does each mechanism control duplicates and misses, and how was that measured? The evidence is mostly count agreement against references that differ between studies (Section 5.6). The 14 comparisons on the same data show that motion-only trackers were not outperformed by trackers with appearance features on look-alike fruit, and that a geometric step removed duplicates that tracking or summation left (Table 5).

How are classes attached to unique fruit? In the 25 studies that report counts per class, the class is mostly carried by the tracker; one study states a voting rule for disagreeing frames (Section 6).

What does the oil-palm literature cover, and what is missing? It covers ripeness classification, detection, and counting in single images or on conveyors. No study associates bunches across views of a palm, and the one dataset with cross-view identity links has been evaluated with statistical baselines only (Section 8).

10.3. Five gaps

(i) Identity is rarely measured. Only 18 of the 149 multi-observation method studies with an abstract (12%) report an identity metric (Section 5.6, Fig. 9). A study that claims to remove duplicates should report, on annotated sequences or view sets, the numbers of duplicate records, missed fruit, false merges of two fruit into one, and identity switches, in addition to the count error. Wang et al. (2019) and Rapado-Rincón et al. (2023) show that this is feasible.

(ii) Mechanisms are rarely compared on the same data. Table 5 contains 14 such comparisons, six of which compare trackers only. Comparisons across mechanisms, for example tracking against geometric association on the same trees, are rare, and comparisons against a calibrated statistical-correction baseline are rarer, although image counts can be corrected to total fruit load with an occlusion factor estimated by regression between manual and machine-vision counts of a set of calibration trees in each orchard (Koirala et al., 2019; Anderson et al., 2021b). A new mechanism should be tested against a statistical correction fitted on separate trees.

(iii) Class-wise identity is almost absent. 25 multi-observation studies report counts per class, and in 2 of them the source states how label disagreement between frames is resolved (Section 6). The required measurement is a per-class count error computed on unique fruit, together with a confusion matrix of classes on unique fruit.

(iv) Discrete-view protocols lack mechanisms. Most methods assume video (Fig. 5b and Fig. 7). Census protocols photograph a plant from a few fixed positions, as in oil palm (Indriani et al., 2026) or in multi-sided mango and citrus imaging (Nayak Nenavath and Perumal, 2024; Zhu et al., 2024). For such data, temporal tracking does not apply, and the evidence on appearance matching, geometric association, and learned association comes mostly from other acquisition designs (Section 4). Wide-baseline association with known or estimated pose is the missing experiment.

(v) Oil palm has detection and ripeness evidence but no identity evidence. A bunch-level census from images needs an evaluation on palms photographed from several sides, with identity links, class labels in census classes, and a per-class count reference (Section 8). One dataset now provides this (Indriani et al., 2026).

10.4. A minimum reporting set

At minimum, a study that claims a duplicate-aware, class-wise inventory should report: (a) the acquisition protocol (views per plant, ordering, pose source, depth source, baseline); (b) the mechanism and the component that removes duplicates; (c) the reference type (hand count on the tree, harvest, or annotated images); (d) the count error per plant and per class; (e) identity metrics on annotated data; and (f) a statistical-correction baseline fitted on separate plants.

11. Limitations of this review

Seven limitations apply. First, Scopus was the only database; work indexed only in Web of Science, IEEE Xplore, or preprint servers is missing unless Scopus also indexes it, one known review that fell outside the year limit of the queries had to be added by hand, and four eligible studies were found only through reference lists. Second, abstracts were not available from the database export used and were retrieved from other bibliographic services; 254 records without an abstract were first decided from titles, sources, and machine-generated summaries.

Third, screening and coding were done by one reviewer working with a large language model. The second pass, the adjudication, and the challenge of each proposed change (Section 2.4) were also carried out by the model. They show how reproducible the decisions are, and they changed 16 group decisions, but they do not replace an independent second reviewer. The first author's blind samples give a human check, with only moderate agreement ($\kappa = 0.57$ for titles and 0.45 for eligibility codes). Most disagreements in those samples were also resolved with the model, and no second person screened a sample independently. The reviewers also delegated to the model the remaining verification decisions: the coding rules in Section 2.5, the blind recoding of the multi-observation and oil-palm studies, the reasons for exclusion that were in question, and the additions from reference lists. Such a check remains the first step needed to strengthen the map. Fourth, the title stage was the least reliable step: in the sample of 300 titles, 11 records that the first pass had excluded met an inclusion criterion once their abstracts were read, and although the exclusions from the two core query sets were re-read from their abstracts, the map is likely to miss eligible studies among the remaining title exclusions, mainly in the supporting groups. Fifth, rule-based codes for crop, modality, platform, and metric types were spot-checked but not verified for every record, and metric types were read only from abstracts, so a study may report a metric in its full text that is not counted here. Full text was obtained for about half of the included studies (476 of 1,013), and the codes of 4 of the 199 multi-observation studies identified through Scopus rest on the abstract alone.

Sixth, the set of methods papers from outside agriculture is purposive, limited to the most cited records of each search. Seventh, results were not pooled, and the numbers in Tables 4 and 5 should be compared only within a row.

12. Conclusions

This review mapped image-based fruit counting published between 2012 and 2026. Reporting followed PRISMA 2020 where its items apply. The review included 1,018 studies, and the 199 multi-observation studies identified through Scopus were coded by association mechanism and acquisition design (Figs. 5–8; Appendix Table A1).

Overall, the literature shows that counting fruit from images is an identity problem as much as a detection problem. Statistical correction is repeatable but must be refitted

and cannot name duplicates. Temporal tracking dominates recent work, and on look-alike fruit, motion-only trackers were not outperformed by trackers with appearance features in the two comparisons made on the same data. Geometric association gives calibration-free counts when pose, scale, and depth are reliable, at the cost of sensors or reconstruction time. Learned association is recent and limited by the lack of identity-labelled data. At the same time, the evidence is unbalanced: 18 of the 149 abstracts report an identity metric, 25 of the 195 method studies report counts per class, Table 5 holds 14 comparisons on the same data, and no oil-palm study associates bunches across views of a palm.

For video along a row, motion-based tracking is the tested default and needs a separate step for fruit seen from both sides. For a few discrete views around a plant, temporal tracking does not apply; association must rest on geometry, on appearance combined with position, or on learned correspondence, and this case is the least tested. For class-wise inventories, the class must be decided per unique fruit, and the rule for disagreeing views should be reported.

Looking forward, the synthesis supports five priorities: (i) measure identity errors, not only the count error; (ii) compare mechanisms on the same data, including a statistical-correction baseline fitted on separate plants; (iii) report per-class counts and class confusion on unique fruit; (iv) test wide-baseline association on discrete-view protocols; and (v) evaluate bunch-level, class-wise counting on oil palms photographed from several sides.

Data availability

The search strings are listed in Appendix B. The screening decision for every record, the coded evidence matrix, the log of changes made during verification, and the analysis scripts that reproduce the numbers and figures are provided as supplementary material.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model (Claude, Anthropic) to read titles, abstracts, and available full texts during screening, to record screening decisions and codes in two separate passes, to decide the verification items that the reviewers delegated to it, to check the statements of this article against the cited sources, to write the analysis scripts, and to draft the text. The authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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A. Coding records of the multi-observation studies

Table A1 lists every multi-observation study included in the review, in order of publication year. The result reported by each study is given in the supplementary evidence matrix.

Table A1
Detailed coding records of the 199 multi-observation studies.

Study	Crop	Platform	Acquisition	Association mechanism	Per-class counts	Metrics in abstract	Coded from
Kassim et al. (2012)	Oil palm	Not stated	Revisits	Temporal tracking	No	No abstract	Full text
Qian et al. (2013)	Apple	Not stated	Discrete views	Statistical correction	No	No abstract	Full text
Wang et al. (2013)	Apple	Ground vehicle	Video	Geometric or 3D association	No	None	Full text
Gongal et al. (2014)	Apple	Not stated	Discrete views	Geometric or 3D association	No	No abstract	Full text
Nuske et al. (2014b)	Grape	Ground vehicle	Video	Geometric or 3D association	No	None	Full text
Nuske et al. (2014a)	Grape	Ground vehicle	Video	Geometric or 3D association	No	None	Full text
Song et al. (2014)	Pepper	Not stated	Video	Geometric or 3D association	No	None	Full text
Linker et al. (2015)	Apple	Not stated	Discrete views	Statistical correction	No	None	Full text
Gongal et al. (2016)	Apple	Not stated	Discrete views	Geometric or 3D association	No	None	Full text
Roy and Isler (2016)	Apple	Not stated	3D scan	Tracking + geometric	No	None	Full text
Stein et al. (2016)	Mango	Not stated	Video	Tracking + geometric	No	Count, detection	Full text
Roy and Isler (2017a)	Apple	Ground vehicle	Discrete views	Geometric or 3D association	No	None	Full text
Roy and Isler (2017b)	Apple	Not stated	Video	Geometric or 3D association	No	No abstract	Full text
Liu et al. (2018)	Not stated	Not stated	Video	Tracking + geometric	No	Count	Full text
Huang and Lin (2019)	Not stated	Ground vehicle	Video	Tracking + geometric	No	None	Full text
Liu et al. (2019)	Mango	Not stated	Video	Tracking + geometric	No	Detection	Full text
Nellithimaru and Kantor (2019)	Apple	Not stated	3D scan	Geometric or 3D association	No	Detection	Full text
Roy et al. (2019)	Apple	Not stated	3D scan	Tracking + geometric	No	None	Full text
Wang et al. (2019)	Mango	Not stated	Video	Temporal tracking	No	Count	Full text
Apolo-Apolo et al. (2020b)	Apple	UAV, Ground vehicle	Discrete views	Geometric or 3D association	No	Count	Full text
Dong et al. (2020)	Not stated	Not stated	3D scan	Geometric or 3D association	No	None	Full text
Gené-Mola et al. (2020a)	Apple	Not stated	3D scan	Geometric or 3D association	No	Count	Full text
Gené-Mola et al. (2020c)	Apple	Not stated	Discrete views	Geometric or 3D association	No	Detection	Full text
Gené-Mola et al. (2020d)	Apple	Not stated	Discrete views	Dataset paper	No	None	Full text
Gené-Mola et al. (2020b)	Apple	Not stated	3D scan	Dataset paper	No	None	Full text
Häni et al. (2020)	Apple	Not stated	Video	Geometric or 3D association	No	None	Full text
Santos et al. (2020)	Grape	Not stated	Video	Geometric or 3D association	No	Detection	Full text
Serafino et al. (2020)	Citrus	Ground vehicle	Video	Temporal tracking	No	None	Full text
Sun et al. (2020)	Cotton	Not stated	3D scan	Geometric or 3D association	No	Count	Full text
Aji et al. (2021)	Oil palm	Laboratory	Video	Temporal tracking	No	None	Full text
Anderson et al. (2021a)	Mango	Not stated	Video	Statistical + tracking	No	Count	Full text
Gao et al. (2021b)	Apple	Not stated	Video	Temporal tracking	No	No abstract	Full text
Gao et al. (2021a)	Apple	Not stated	Video	Temporal tracking	No	Detection	Full text
Itakura et al. (2021)	Apple	Not stated	Video	Temporal tracking	No	Detection	Full text
Kirk et al. (2021)	Not stated	Not stated	Video	Appearance + tracking	Yes	No abstract	Full text
Osman et al. (2021a)	Apple	Ground vehicle	Video	Appearance + tracking	No	Detection	Full text
Osman et al. (2021b)	Apple	Not stated	Video	Appearance + tracking	No	Detection	Full text
Parico and Ahamed (2021)	Pear	Not stated	Video	Appearance + tracking	No	Detection	Full text
Scalisi et al. (2021)	Apple	Not stated	Video	Statistical correction	No	No abstract	Full text
Seo et al. (2021)	Tomato	Ground vehicle	Video	Temporal tracking	Yes	None	Full text
Torres-Sánchez et al. (2021)	Grape	UAV, Ground vehicle	3D scan	Geometric or 3D association	No	Count	Full text

Study	Crop	Platform	Acquisition	Association mechanism	Per-class counts	Metrics in abstract	Coded from
Zaenker et al. (2021)	Pepper	Ground vehicle	3D scan	Geometric or 3D association	No	None	Full text
Zine-El-Abidine et al. (2021)	Apple	Not stated	3D scan	Geometric or 3D association	No	None	Full text
Dai et al. (2022)	Tomato	Ground vehicle	Video	Tracking + geometric	No	No abstract	Full text
Daud et al. (2022)	Oil palm	Not stated	Video	No association + tracking	No	None	Full text
Egi et al. (2022)	Tomato	UAV	Video	Appearance + tracking	Yes	Detection	Full text
Gao et al. (2022)	Apple	Not stated	Video	Temporal tracking	No	No abstract	Full text
Ge et al. (2022)	Tomato	Ground vehicle	Video	Appearance + tracking	No	Detection	Full text
He et al. (2022b)	Not stated	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Marangoz et al. (2022)	Pepper	Ground vehicle	3D scan	Geometric or 3D association	No	Detection	Full text
Turečková et al. (2022)	Tomato	Not stated	Video	Geometric or 3D association	No	No abstract	Full text
Wang et al. (2022)	Apple	UAV	Video	Temporal tracking	No	None	Full text
Xia et al. (2022)	Apple	Not stated	Video	Appearance + learned	No	Count, detection	Full text
Zhang et al. (2022a)	Citrus	Not stated	Video	Temporal tracking	No	Count, detection	Full text
Zhang et al. (2022b)	Apple	Not stated	Discrete views	Statistical correction	No	Count	Full text
Zhu et al. (2022)	Citrus	Not stated	3D scan	Geometric or 3D association	No	No abstract	Full text
Abeyrathna et al. (2023)	Apple	Ground vehicle	Video	Appearance + tracking	No	Count, detection	Full text
Ariza-Sentís et al. (2023b)	Grape	UAV, Ground vehicle	Video	Dataset paper	No	None	Full text
Ariza-Sentís et al. (2023a)	Grape	UAV, Ground vehicle	Video	Tracking + learned	No	None	Full text
Arlotta et al. (2023)	Grape	Ground vehicle, Laboratory	Video	Tracking + geometric	No	None	Full text
Ciarfuglia et al. (2023)	Grape	Ground vehicle	Video	Appearance + tracking + geometric	No	None	Full text
Daud et al. (2023)	Oil palm	Not stated	Video	Temporal tracking	No	No abstract	Full text
Gené-Mola et al. (2023)	Apple	Not stated	Video	Appearance + tracking	No	Count, identity, detection	Full text
Gremes et al. (2023)	Citrus	Ground vehicle	Video	Temporal tracking	No	Count, detection	Full text
Guo et al. (2023)	Kiwifruit	Not stated	Video	Appearance + tracking	No	No abstract	Abstract
Hu et al. (2023)	Apple	Not stated	Video	Appearance + tracking	No	Count, detection	Full text
Lawal et al. (2023)	Strawberry	Not stated	Video	Temporal tracking	No	Detection	Full text
Li et al. (2023)	Blueberry	Ground vehicle	Discrete views	Statistical correction	No	Count	Full text
Lin et al. (2023)	Citrus	Not stated	Video	Appearance + tracking	No	Count	Full text
Lyu et al. (2023)	Grape	Not stated	Video	Temporal tracking	No	No abstract	Full text
Parico and Ahamed (2023)	Pear	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Qureshi et al. (2023)	Apple	Ground vehicle	3D scan	Geometric or 3D association	No	Count, detection	Full text
Rapado-Rincón et al. (2023)	Tomato	Ground vehicle	3D scan	Tracking + geometric	No	None	Full text
Shen et al. (2023)	Grape	Not stated	Video	Temporal tracking	No	No abstract	Full text
Shiddiq (2023)	Oil palm	Laboratory	Video	Temporal tracking	No	Count	Full text
Vijayakumar et al. (2023)	Citrus	UAV	Discrete views	Statistical correction	No	Count	Full text
Villacrés et al. (2023)	Apple	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Wu et al. (2023)	Apple	Not stated	Video	Temporal tracking	No	Count, identity, detection	Full text
Xie et al. (2023)	Not stated	Ground vehicle	Discrete views	Geometric or 3D association	No	None	Full text
Zaenker et al. (2023)	Pepper	Ground vehicle	3D scan	Geometric or 3D association	No	None	Full text
Zheng et al. (2023a)	Strawberry	Not stated	Discrete views	Geometric + learned	Yes	None	Full text
Zheng et al. (2023b)	Citrus	UAV, Ground vehicle	Video	Appearance + tracking	No	Count, detection	Full text
Zhou et al. (2023)	Strawberry	Not stated	Video	Temporal tracking	Yes	Count	Full text
Ahmedt-Aristizabal et al. (2024)	Grape	Not stated	Video	Appearance + tracking	No	None	Full text
An et al. (2024)	Strawberry	Not stated	Video	Appearance + tracking	Yes	No abstract	Full text
Ariza-Sentís et al. (2024)	Grape	UAV	Video	Dataset paper	No	None	Full text
Ashokan and Akbar (2024)	Not stated	Not stated	Video	Appearance + tracking	No	None	Full text
Du et al. (2024)	Pepper	Not stated	Video	Appearance + tracking	No	Count, identity, detection	Full text
Feng et al. (2024)	Citrus	Not stated	Video	Appearance + tracking	No	Count	Full text

Study	Crop	Platform	Acquisition	Association mechanism	Per-class counts	Metrics in abstract	Coded from
Garderes et al. (2024)	Apple	Not stated	Video	Appearance + tracking	No	Identity, detection	Full text
Hernandez and Medeiros (2024)	Not stated	Not stated	Video	Tracking + geometric	No	No abstract	Full text
Ishizuka et al. (2024)	Not stated	Not stated	Video	Temporal tracking	No	None	Full text
Kang et al. (2024)	Tomato	Ground vehicle, Handheld	Video	Temporal tracking	No	Count, detection	Full text
Magalhães et al. (2024)	Tomato	Laboratory	Discrete views	Geometric or 3D association	No	Count	Full text
Matos et al. (2024)	Apple	Ground vehicle	Video	Geometric or 3D association	No	None	Full text
Meyer et al. (2024)	Apple	Not stated	3D scan	Geometric or 3D association	No	None	Full text
Muzaddid and Beksi (2024)	Cotton	UAV, Ground vehicle	Video	Temporal tracking	No	None	Full text
Nayak Nenavath and Perumal (2024)	Mango	Not stated	Discrete views	Statistical correction	No	Count	Full text
Pai and Lin (2024)	Tomato	Not stated	Video	Appearance + tracking	Yes	None	Full text
Qi et al. (2024)	Tomato	Not stated	Video	Appearance + tracking	Yes	No abstract	Full text
Safre et al. (2024)	Cherry	Not stated	Video	Appearance + tracking	No	No abstract	Abstract
Santos et al. (2024)	Citrus	Not stated	Video	Tracking + geometric	No	No abstract	Full text
Saraceni et al. (2024)	Grape	Ground vehicle	Video	Temporal tracking	No	None	Full text
Scalisi et al. (2024)	Pear	Not stated	Video	Statistical correction	Yes	No abstract	Full text
Sekharamantry et al. (2024)	Apple	UAV	Video	Temporal tracking	No	Count, detection	Full text
Tu et al. (2024)	Passion fruit	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Villacrés and Vougioukas (2024)	Apple	Ground vehicle	Video	Geometric or 3D association	No	No abstract	Full text
Viveros Escamilla et al. (2024)	Citrus	Not stated	Video	Appearance + tracking	No	None	Full text
Wang et al. (2024a)	Pitaya	Not stated	3D scan	Tracking + geometric	No	No abstract	Full text
Wang et al. (2024b)	Camellia	Not stated	Video	Appearance + tracking	Yes	No abstract	Full text
Xia et al. (2024)	Kiwifruit	Not stated	Video	Temporal tracking	No	No abstract	Full text
Xiao et al. (2024)	Cotton	Not stated	3D scan	Geometric or 3D association	No	No abstract	Full text
Yang et al. (2024b)	Apple	Not stated	Video	Appearance + tracking	No	Count, identity, detection	Full text
Yuan et al. (2024)	Strawberry	Not stated	Video	Geometric or 3D association	No	Count, detection	Full text
Zhang et al. (2024)	Peach	Ground vehicle	Video	Temporal tracking	No	Count, identity	Full text
Zheng et al. (2024)	Citrus	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Zhou et al. (2024a)	Citrus	Not stated	3D scan	Geometric or 3D association	No	Count	Full text
Zhou et al. (2024b)	Strawberry	Not stated	Video	Appearance + tracking	Yes	Count	Full text
Zhu et al. (2024)	Citrus	Not stated	Discrete views	Statistical correction	No	None	Full text
Akella et al. (2025)	Coconut	Not stated	Video	Temporal tracking	No	Detection	Full text
Ariza-Sentís et al. (2025)	Grape	UAV	Video	Tracking + learned	No	Count	Full text
Cao et al. (2025)	Apple	UAV	Video	Appearance + tracking	No	Count, identity, detection	Full text
Cho et al. (2025)	Not stated	Laboratory	3D scan	Geometric or 3D association	No	Count	Full text
Dong et al. (2025)	Camellia	Not stated	Video	Appearance + tracking	No	Detection	Full text
Garg et al. (2025)	Not stated	UAV, Ground vehicle	3D scan	Tracking + geometric	Yes	Count	Full text
Hassan et al. (2025)	Citrus	Handheld	3D scan	Appearance + tracking + geometric	No	None	Full text
Hobart et al. (2025)	Apple	UAV	3D scan	Geometric or 3D association	No	None	Full text
Hong et al. (2025)	Persimmon	Not stated	Discrete views	Geometric or 3D association	No	Count	Full text
Isobe et al. (2025)	Citrus	Not stated	Video	Statistical correction	No	None	Full text
Issad et al. (2025)	Not stated	Not stated	Video	Temporal tracking	No	Detection	Full text
Jiang et al. (2025)	Cotton	Ground vehicle, Handheld	3D scan	Geometric or 3D association	No	Count, detection	Full text
Kemper et al. (2025)	Grape	Not stated	Video	Appearance + tracking	No	Count	Full text
Kutyrev et al. (2025)	Apple	UAV, Ground vehicle	Discrete views	Geometric or 3D association	No	Detection	Full text
Lei et al. (2025)	Apple	Not stated	Revisits	Appearance + tracking + geometric	No	Count	Full text
Louis et al. (2025)	Melon	Ground vehicle, Laboratory	Video	Temporal tracking	No	Count	Full text

Study	Crop	Platform	Acquisition	Association mechanism	Per-class counts	Metrics in abstract	Coded from
Lu et al. (2025)	Litchi	Not stated	Video	Temporal tracking	No	Count	Full text
Lyu et al. (2025a)	Grape	Not stated	Video	Temporal tracking	No	No abstract	Full text
Lyu et al. (2025b)	Grape	Not stated	Video	Temporal tracking	No	No abstract	Full text
Majdalawieh and Khan (2025)	Not stated	Not stated	Video	Appearance + tracking	No	Detection	Full text
Matos et al. (2025)	Apple	Not stated	3D scan	Geometric or 3D association	No	No abstract	Full text
Mendez et al. (2025)	Pepper	Ground vehicle	Video	Appearance + tracking	No	Identity, detection	Full text
Meyer et al. (2025)	Apple	Not stated	3D scan	Geometric + learned	No	None	Full text
Mollineda et al. (2025)	Citrus	Ground vehicle	Video	Statistical correction	No	None	Full text
Ogundele et al. (2025)	Strawberry	Not stated	Video	Temporal tracking	Yes	Count, detection	Full text
Pichhika et al. (2025)	Mango	Not stated	Video	Temporal tracking	No	No abstract	Full text
Ponce-Machete and Fajardo (2025)	Not stated	UAV	Video	Appearance + tracking	No	Identity, detection	Full text
Qi et al. (2025)	Tomato	Not stated	Video	Appearance + tracking	No	Count, identity, detection	Full text
Sasse et al. (2025)	Grape	Handheld	Video	Temporal tracking	No	Detection	Full text
Singh et al. (2025)	Strawberry	UAV	Video	Temporal tracking	No	Detection	Full text
Tan et al. (2025b)	Apple	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Timprae et al. (2025)	Tomato	Not stated	3D scan	Geometric or 3D association	No	None	Full text
Tu et al. (2025)	Passion fruit	Not stated	Video	Temporal tracking	No	No abstract	Full text
Vaishnavi et al. (2025)	Citrus	Not stated	Video	Appearance + tracking	Yes	Count	Full text
Wang et al. (2025a)	Tomato	Not stated	Video	Temporal tracking	Yes	Count	Full text
Wang et al. (2025b)	Tomato	Ground vehicle	Video	Geometric or 3D association	No	Count, detection	Full text
Wei et al. (2025)	Apple	Not stated	Video	Tracking + geometric	No	Count, identity	Full text
Wu et al. (2025)	Strawberry	Not stated	Video	Temporal tracking	Yes	Count, detection	Full text
Xie-Li and Fallas-Moya (2025)	Pineapple	UAV	Video	Temporal tracking	No	Identity, detection	Full text
Xing et al. (2025)	Tomato	Ground vehicle	Video	Appearance + tracking	Yes	Identity, detection	Full text
Yan et al. (2025)	Apple	Not stated	Video	Appearance + tracking	No	Detection	Full text
Yang et al. (2025a)	Not stated	UAV, Ground vehicle	Video	Tracking + geometric	No	None	Full text
Yang et al. (2025b)	Not stated	UAV, Ground vehicle, Laboratory	Video	Geometric or 3D association	No	None	Full text
Zhai et al. (2025)	Not stated	Not stated	Video	Appearance + tracking	No	Count, identity	Full text
Zhang et al. (2025)	Kiwifruit	Handheld	Video	Appearance + tracking	No	No abstract	Full text
Zhang and Cao (2025)	Blueberry	Not stated	Video	Temporal tracking	No	Detection	Full text
Zheng et al. (2025)	Tomato	Ground vehicle	Video	Tracking + geometric	Yes	Identity	Full text
Zhou et al. (2025)	Banana	Not stated	Video	Appearance + tracking	No	None	Full text
Awal et al. (2026)	Citrus	Ground vehicle	Video	Temporal tracking	Yes	Count, detection	Full text
Bharti et al. (2026)	Apple	Not stated	3D scan	Tracking + geometric	No	Detection	Full text
Blondet et al. (2026)	Avocado	Not stated	Video	Appearance + tracking	No	No abstract	Full text
Farhoud et al. (2026)	Citrus	Not stated	Video	Temporal tracking	Yes	Count, detection	Full text
Felip-Pomés et al. (2026)	Apple	Not stated	Video	Temporal tracking	No	Count, detection	Full text
Feng et al. (2026)	Tomato	Not stated	Video	Temporal tracking	No	No abstract	Full text
Fusaro et al. (2026)	Apple	Ground vehicle	Revisits	Appearance + geometric + learned	No	None	Full text
Gao et al. (2026)	Not stated	Ground vehicle	Video	Tracking + geometric	No	None	Full text
Hu et al. (2026)	Strawberry	Not stated	Discrete views	Geometric or 3D association	No	No abstract	Full text
Huang et al. (2026)	Mango	Not stated	Video	Appearance + tracking	Yes	No abstract	Full text
Indriani et al. (2026)	Oil palm	Handheld	Discrete views	Statistical correction	Yes	None	Full text
Jiang et al. (2026)	Cotton	Ground vehicle	Video	Geometric or 3D association	No	No abstract	Full text
Lee et al. (2026)	Pear	Not stated	3D scan	Geometric or 3D association	No	Count, detection	Full text
Lin et al. (2026)	Tomato	Not stated	Video	Appearance + tracking	Yes	None	Full text
Nguyen et al. (2026)	Apple	UAV, Ground vehicle, Laboratory	3D scan	Geometric or 3D association	No	Count	Full text
Pardo-Beainy et al. (2026)	Strawberry	Not stated	Video	Appearance + tracking	Yes	Count	Full text
Park et al. (2026)	Not stated	Not stated	Video	Geometric or 3D association	No	No abstract	Full text

Study	Crop	Platform	Acquisition	Association mechanism	Per-class counts	Metrics in abstract	Coded from
Piazolo et al. (2026)	Grape	UAV, Ground vehicle	Video	Appearance + tracking	No	Count, identity	Full text
Santos (2026)	Not stated	Ground vehicle	Video	Tracking + geometric + learned	No	None	Full text
Si et al. (2026)	Citrus	Handheld	Video	Appearance + tracking	No	Count, identity, detection	Full text
Tu et al. (2026)	Passion fruit	Not stated	Video	Temporal tracking	No	No abstract	Abstract
Wang et al. (2026a)	Apple	UAV, Laboratory	3D scan	Tracking + geometric	No	None	Full text
Wang et al. (2026b)	Not stated	UAV, Ground vehicle	Discrete views	Geometric or 3D association	No	Count	Full text
Weng et al. (2026)	Citrus	UAV	Video	Temporal tracking	No	No abstract	Full text
Woo et al. (2026)	Grape	Handheld	Video	Geometric or 3D association	No	Detection	Full text
Xing et al. (2026)	Tomato	Ground vehicle	Video	Appearance + tracking	Yes	Count	Full text
Yang et al. (2026)	Grape	Not stated	Video	Geometric or 3D association	No	Count, detection	Full text
Yok-In et al. (2026)	Grape	Ground vehicle	3D scan	Geometric or 3D association	No	Count, detection	Abstract
Yoshida et al. (2026)	Grape	Ground vehicle	Revisits	Geometric or 3D association	No	None	Full text
Zhao et al. (2026a)	Not stated	Not stated	3D scan	Geometric or 3D association	No	None	Full text
Zhao et al. (2026b)	Apple	Ground vehicle	3D scan	Geometric + learned	No	Detection	Full text
Zhu et al. (2026)	Not stated	Ground vehicle	Video	Appearance + tracking	No	Identity	Full text

Note: A plus sign joins the mechanisms that a study combines. Not stated, the property is not stated in the abstract. Metric types were read from abstracts only: Count, agreement between estimated and reference counts; Identity, a tracking metric such as MOTA, IDF1, HOTA, or ID switches; Detection, detection quality; No abstract, no abstract was available. Coded from: the most complete source that was read when the codes were assigned. UAV, unmanned aerial vehicle.

B. Scopus search strings

Each string was submitted to the Scopus advanced search on 28 September 2026 (UTC). The keys Q1 to Q9 are those of Table 1. The searches under Q8a–Q8e and Q9 were sorted by citation count.

Q1. Multi-observation counting (871 records)

```
TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (count* OR "yield estimat*" OR "yield predict*" OR "yield forecast*" OR "load estimat*" OR "fruit load" OR "crop load" OR inventor* OR enumerat* OR "fruit number") AND ("multi-view" OR multiview OR "multiple view*" OR "multi-camera" OR "multiple camera*" OR video* OR "image sequence*" OR track* OR "structure from motion" OR "structure-from-motion" OR SfM OR SLAM OR "3D reconstruct*" OR "point cloud*" OR "double count*" OR duplicat* OR "re-identification" OR "data association" OR "cross-view" OR "both sides" OR "two sides") AND (image* OR camera* OR vision OR "deep learning" OR "neural network*" OR detect* OR segment* OR YOLO OR "R-CNN" OR photograph*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")
```

Q2. Counting in titles (757 records)

```
TITLE((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (count* OR "yield estimat*" OR "yield predict*" OR "load estimat*" OR "fruit load" OR "crop load" OR "fruit number")) AND TITLE-ABS-KEY((image* OR camera* OR vision OR "deep learning" OR "neural network*" OR detect* OR segment* OR YOLO OR "R-CNN" OR photograph*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")
```

Q3. Oil-palm imaging (1,079 records)

```
TITLE-ABS-KEY(("oil palm" OR "Elaeis guineensis" OR "fresh fruit bunch*" OR "palm oil fruit*") AND (image* OR vision OR camera* OR "deep learning" OR "machine learning" OR detect* OR convolutional OR YOLO OR segment* OR classif* OR ripeness OR maturity OR count*) AND (bunch* OR fruit* OR ripeness OR maturity OR harvest*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC))
```

Q4. Class attributes with counting (361 records)

```
TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (ripeness OR maturity OR "ripening stage*" OR "maturity stage*" OR "size class*" OR grade* OR grading OR defect* OR "multi-class" OR "per class" OR "class-wise") AND (count* OR "yield estimat*" OR inventor* OR "load estimat*") AND (detect* OR track* OR segment*) AND (image* OR camera* OR video* OR vision)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC)) AND NOT
TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")
```

Q5. Depth and 3D sensing (1,988 records)

```
TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date
```

palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND ("RGB-D" OR RGBD OR "depth camera*" OR "depth image*" OR "depth map*" OR "depth sensor*" OR "depth information" OR "stereo camera*" OR "stereo vision" OR binocular OR LiDAR OR "time-of-flight" OR "point cloud*" OR Kinect OR RealSense OR "3D camera*") AND (detect* OR count* OR locali* OR "yield estimat*" OR "size estimat*") AND (orchard* OR field* OR tree* OR canopy OR plant* OR greenhouse* OR vineyard*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q6. Reviews of association methods (236 records)

TITLE-ABS-KEY(("multi-object tracking" OR "multiple object tracking" OR "multi-target tracking" OR "multiple target tracking" OR "re-identification" OR "multi-camera tracking" OR "multi-camera multi-object" OR "multi-view detection" OR "multiview detection" OR "cross-view association" OR "multi-view association" OR "cross-camera association" OR "object counting" OR "structure from motion" OR "visual SLAM" OR "instance association") AND (review OR survey OR benchmark)) AND DOCTYPE(re) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI))

Q7. Reviews of fruit detection and yield (1,049 records)

TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (count* OR "yield estimat*" OR "yield predict*" OR "load estimat*" OR detect* OR ripeness OR maturity OR harvest* OR locali*) AND (image* OR vision OR camera* OR sensor* OR "deep learning" OR "machine learning")) AND DOCTYPE(re) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q8a. Methods outside agriculture: multi-object tracking (3,604 records; the 25 most cited were retained)

TITLE("multi-object tracking" OR "multiple object tracking" OR "multi-target tracking" OR "tracking-by-detection" OR "tracking by detection") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8b. Methods outside agriculture: multi-view detection and association (2,243 records; the 25 most cited were retained)

TITLE("multi-view" OR multiview OR "multi-camera" OR "multiple cameras" OR "cross-view") AND TITLE(detection OR tracking OR association OR counting OR "re-identification") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8c. Methods outside agriculture: structure from motion and SLAM (9,696 records; the 25 most cited were retained)

TITLE("structure from motion" OR "structure-from-motion" OR SLAM OR "bundle adjustment" OR "multi-view stereo") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8d. Methods outside agriculture: object counting (1,554 records; the 25 most cited were retained)

TITLE("object counting" OR "crowd counting" OR "counting objects" OR "class-agnostic counting") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8e. Methods outside agriculture: re-identification (6,435 records; the 25 most cited were retained)

TITLE("re-identification" OR "reidentification") AND TITLE(object OR vehicle OR person OR instance) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q9. Named foundational methods (1,082 records; the 25 most cited were retained)

(TITLE("simple online and realtime tracking") OR TITLE("deep association metric") OR TITLE("BoT-SORT") OR TITLE("StrongSORT") OR TITLE("multiview detection with feature perspective transformation") OR TITLE("representing scenes as neural radiance fields") OR TITLE("3D Gaussian splatting for real-time radiance field rendering") OR TITLE("depth anything") OR TITLE("segment anything") OR TITLE("vision transformers for dense prediction") OR TITLE("end-to-end object detection with transformers") OR TITLE("DETRs beat YOLOs")) AND PUBYEAR > 2011 AND PUBYEAR < 2027